

1.1. APACHE NIFI KASU PRAKTIKOAK

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Apache NiFi instalazioa (Docker bidez)

1. Docker Desktop abiarazi (*engine running*)
2. PowerShell-en administratzaile gisa, komando hau exekutatu:

```
docker run --name nifi -p 8443:8443 -d -e SINGLE_USER_CREDENTIALS_USERNAME=nifi -e SINGLE_USER_CREDENTIALS_PASSWORD=nifinifinifi -e NIFI_JVM_HEAP_MAX=2g -v "C:\nifi\ariketak:/opt/nifi/ariketak" apache/nifi:latest
```

```
PS C:\WINDOWS\system32> docker run --name nifi -p 8443:8443 -d -e SINGLE_USER_CREDENTIALS_USERNAME=nifi -e SINGLE_USER_CREDENTIALS_PASSWORD=nifinifinifi -e NIFI_JVM_HEAP_MAX=2g apache/nifi:latest
Unable to find image 'apache/nifi:latest' locally
latest: Pulling from apache/nifi
4f4fb70def54: Pull complete
c5d6f1e3a620: Pull complete
906ceac66c1a: Pull complete
33ab2d4f2c9c: Pull complete
29b008335e6f: Pull complete
336e2f4f5fbf: Pull complete
c4013e1e3834: Pull complete
8f8bb581d4de: Pull complete
b692816de839: Pull complete
90858adb9dfc: Pull complete
6d3c20e8e1e8: Pull complete
fe527ec015c4: Download complete
Digest: sha256:7daebe23308be2b68ad8163b1f8f45c37bba506c569a1f9e0de25af9183430ea
Status: Downloaded newer image for apache/nifi:latest
41cfe9531f9ae15dbef48ab921a42f24b2ee138258c51ba3e03edf1e649fdae3
PS C:\WINDOWS\system32>
```

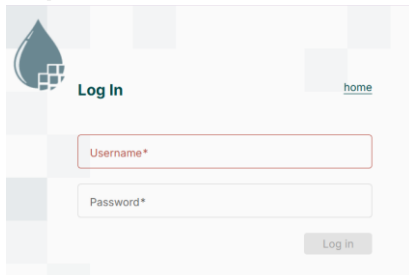
3. PowerShell-en exekutatu:

```
docker ps
```

```
PS C:\WINDOWS\system32> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
41cfe9531f9a   apache/nifi:latest   "../scripts/start.sh"   About a minute ago   Up About a minute   0.0.0.0:8443->8443/tcp, [::]:8443->8443/tcp
PS C:\WINDOWS\system32>
```

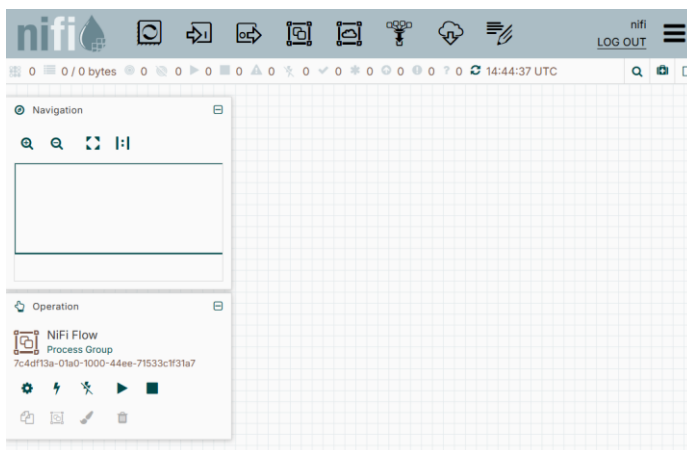
4. Nabigatzailearen bidez ondorengo estekan sartu:

https://localhost:8443/nifi



5. Saioa hasi:

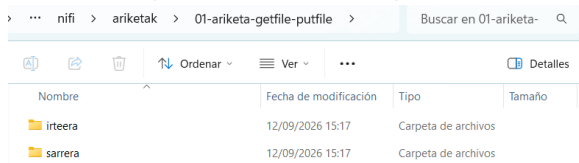
- Username: **nifi**
- Password: **nifinifinifi**



Apache NiFi 1. kasua: Fitxategiak mugitu

Helburua: Fitxategiak sarrera-karpeta batetik irteera-karpeta batera transferitu, gatazka-kudeaketarekin (jadanik izen berdineko fitxategi bat badago irteera-karpetan, fitxategiari izen berria jarri).

1. Sarrera-karpeta eta irteera-karpeta sortu Windows-en.



2. **Process Group berria:**

ProcessGroup arrastatu, ariketa honi dagozkion elementuak bertan sartzeko. Dagokion izena jarri eta **Add** sakatu.



Create Process Group

Name*
NiFi_Flow_01_kasua

Parameter Context

Cancel Add

Sortuta dagoenean, klik bikoitza egin **Process Group**-aren gainean, bertan sartzeko.

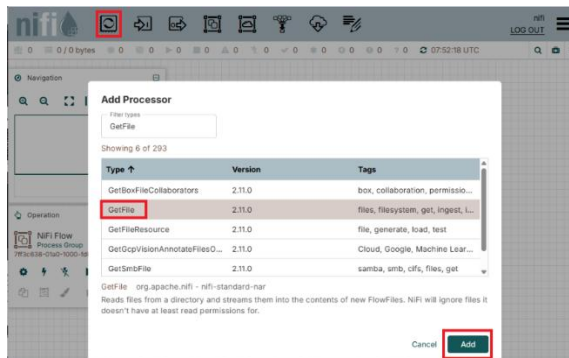
3. **GetFile prozesadorea:**

Gehitu **GetFile** prozesadorea, sarrera-karpetatik fitxategiak eskuratzeko (irakurri eta kendu):

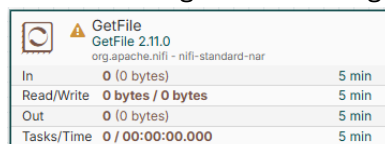
- a. **Processor** arrastatu.



- b. **GetFile** aukeratu eta **Add** sakatu.



- c. Klik bikoitza egin **GetFile**-en gainean, propietateak aldatzeko.



- d. **Properties** atalean:

- **Input Directory:** /opt/nifi/ariketak/01-ariketa-getfile-putfile/sarrera
Docker barruan bide-mapping-ak kontuan hartu behar dira (NiFi kontainerrak es du host-aren / ikusten zuzenean)
- **Keep Source File:** false

Lehenetsia false da, fitxategiak prozesatu ondoren jatorrizko karpetatik kentzeko (produktzio-arrasto OHIKOA da, fitxategi bera birritan ez prozesatzeko). True-rekin gorde egiten da.

- **File Filter: [^\\.*]**

Lehenetsia .* erabiltzen da (fitxategi guztiak). Aukerak:

- .* = guztiak
- *.csv = csv soilik
- ^report.*\\.json = report-z hasten diren json-ak
- ...

- **Minimum File Age: 2 sec**

Fitxategiek behar duten gutxieneko adina, prozesatu ahal izateko. Oso garrantzitsua da. Fitxategi bat idazten ari den unean prozesatzen saiatuz gero, ERDIZKAKO fitxategia jasoko du NiFi-k. 1-2 segundo-tarteak OHIKOAK dira (idazketak amaitu arte denbora emateko).

e. **Settings atalean:**

- **Name: FitxategiaEskuratu**

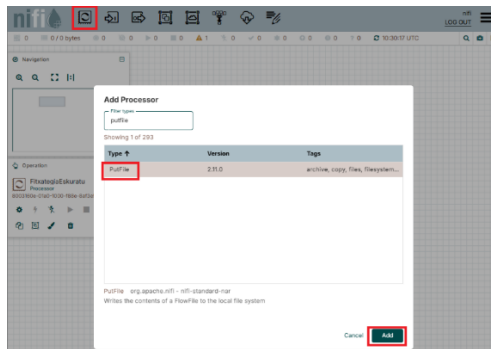
4. **PutFile prozesadorea:**

Gehitu **PutFile** prozesadorea, irteera-karpetan idazteko.

a. **Processor** arrastatu.



b. **PutFile** aukeratu eta **Add** sakatu.

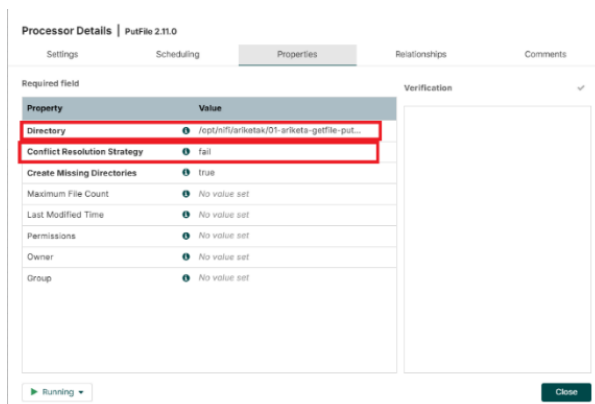


c. Klik bikoitza egin **PutFile**-en gainean, propietateak aldatzeko.

	PutFile PutFile 2.11.0 org.apache.nifi - nifi-standard-nar
In	0 (0 bytes) 5 min
Read/Write	0 bytes / 0 bytes 5 min
Out	0 (0 bytes) 5 min
Tasks/Time	0 / 00:00:00.000 5 min

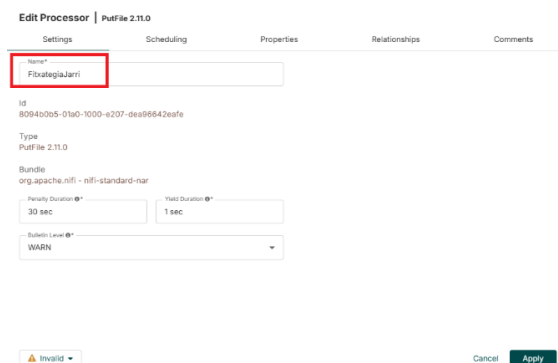
d. **Properties** atalean:

- **Directory:** `/opt/nifi/ariketak/01-ariketa-getfile-putfile/irteera`
Helmuga-karpeta absolutua
- **Conflict Resolution Strategy:** fail
Kritikoa da. Aukerak:
 - FAIL (lehenetsia, errorea botatzen du)
 - REPLACE (gainidatzi)
 - IGNORE (ahaztu, ez idatzi)
 - APPEND (edukia gehitu)



e. **Settings** atalean:

- **Name:** FitxategiaJarri

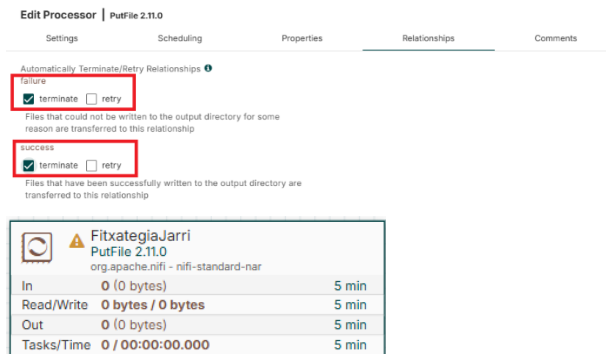


f. **Relationships** atalean:

- **Automatically Terminate/Retry Relationships:**
 - failure: **terminate**
 - success: **terminate**

NiFi-ko prozesadore bakoitzak “irteera-harreman” (relationship) bat edo gehiago ditu (success, failure...). Harreman horiek guztiek “irteera” bat behar dute (beste prozesadorera konektatzeko) edo AUTO-TERMINATE markatuta izan behar dute (NiFi-ri esateko “horiek ahaztuta utzi”).

Harreman bat konektatu gabe edo AUTO-TERMINATE gabe = ‘INVALID’ egoera, prozesadorea ezin abiarazi.



5. **UpdateAttribute prozesadorea:**

Gehitu **UpdateAttribute**, gatazkak kudeatzeko (fitxategi bikoiztuak saihestu).

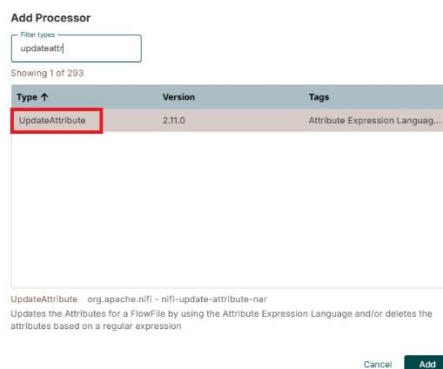
Sortu dugun **PutFile**-aren *failure* harremana bigarren **PutFile** batera bideratuko dugu (/opt/nifi/ariketak/01-ariketa-getfile-putfile/irteera/gatazkak).

UpdateAttribute prozesadorea erabiliko dugu fitxategi-izenak berrizendatzeko:

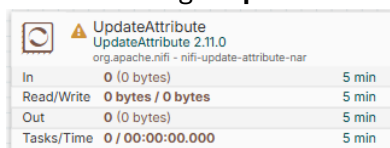
a. **Processor** arrastatu.



b. **UpdateAttribute** aukeratu eta **Add** sakatu.



c. Klik bikoitza egin **UpdateAttribute**-ren gainean, propietateak aldatzeko.



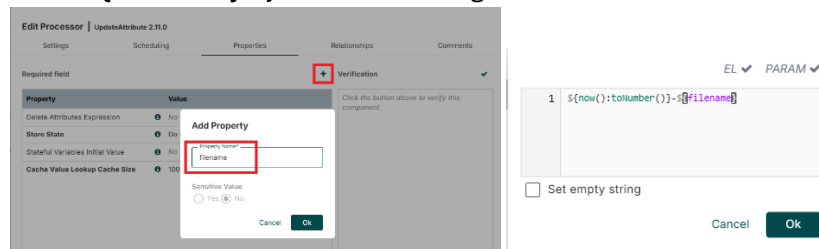
d. **Properties** atalean:

- Propietate berria gehitu **+** ikurra sakatuz: **filename**
- Balioa: **\${now():toNumber()}-\${filename}**

Balio dinamikoa sortuko dugu NiFi Expression Language (NEL)-ekin: `${aldagaia}` sintaxia atributuak irakurtzeko/transformatzeko, funtzioekin: `now()`, `toNumber()`, `substring()`, `replace()`, `format()`...

Balio berriaren azalpena:

- **`${now():toNumber()}`** = une honetako data eta ordua milisegundotan adierazten duen zenbakia. Zenbaki hau une bakoitzean desberdina denez, fitxategi-izen bakarra sortzeko erabil daiteke.
- - karakterea
- **`${filename}`** = jatorrizko fitxategiaren izena



Honela, **PutFile**-k orain izen UNIKOA idatziko du. Beraz, konflikturik ez.

	UpdateAttribute
	UpdateAttribute 2.11.0
	org.apache.nifi - nifi-update-attribute-nar
In	0 (0 bytes) 5 min
Read/Write	0 bytes / 0 bytes 5 min
Out	0 (0 bytes) 5 min
Tasks/Time	0 / 00:00:00.000 5 min

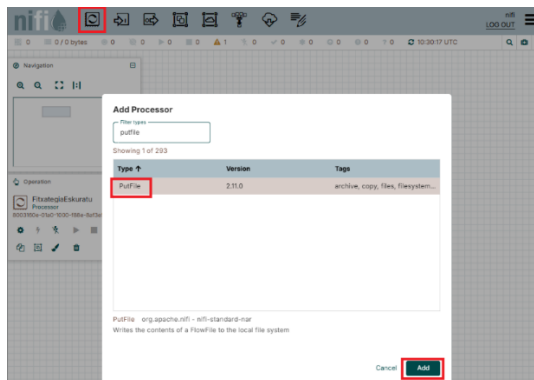
6. Bigarren **PutFile** prozesadorea:

Gehitu bigarren **PutFile** bat, 'gatazkak' karpeta bildu ditzan failure-ak.

- Processor** arrastatu.



- PutFile** aukeratu eta **Add** sakatu.



- Klik bikoitza egin **PutFile**-en gainean, propietateak aldatzeko.

	PutFile
	PutFile 2.11.0
	org.apache.nifi - nifi-standard-nar
In	0 (0 bytes) 5 min
Read/Write	0 bytes / 0 bytes 5 min
Out	0 (0 bytes) 5 min
Tasks/Time	0 / 00:00:00.000 5 min

- Properties** atalean:

- **Directory:** `/opt/nifi/ariketak/01-ariketa-getfile-putfile/irteera/gatazkak`
Helmuga-karpeta absolutua
- **Conflict Resolution Strategy:** fail

Edit Processor | PutFile 2.11.0

Settings Scheduling **Properties** Relationships Comments

Required field + Verification ✓

Property	Value
Directory	/opt/nifi/arketak/01-arketa-getfile-p...
Conflict Resolution Strategy	fail
Create Missing Directories	true
Maximum File Count	No value set
Last Modified Time	No value set
Permissions	No value set

Click the button above to verify this component.

Stopped Cancel Apply

e. **Settings** atalean:

- **Name:** GatazkaFitxategiaMugitu

Edit Processor | PutFile 2.11.0

Settings Scheduling Properties Relationships Comments

Name* GatazkaFitxategiaMugitu

Id 80bab68b-01a0-1000-8564-3600309cDeae

Type PutFile 2.11.0

Bundle org.apache.nifi - nifi-standard-nar

Penalty Duration* 30 sec Yield Duration* 1 sec

Bulletin Level* WARN

f. **Relationships** atalean:

- **Automatically Terminate/Retry Relationships:**
 - failure: **terminate**
 - success: **terminate**

Edit Processor | PutFile 2.11.0

Settings Scheduling Properties **Relationships** Comments

Automatically Terminate/Retry Relationships ⓘ

failure ☒ terminate ☐ retry
Files that could not be written to the output directory for some reason are transferred to this relationship

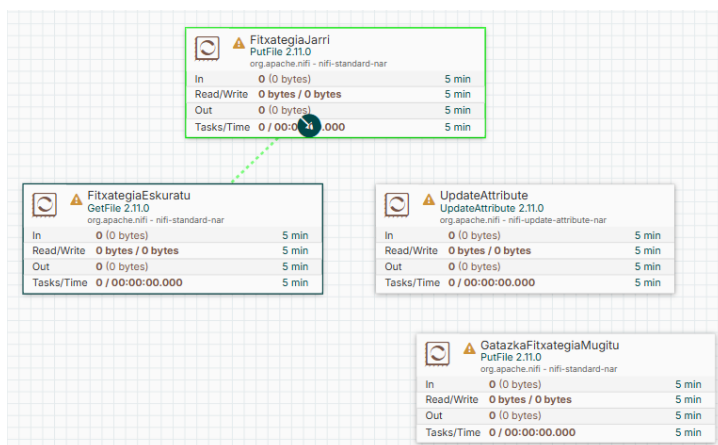
success ☒ terminate ☐ retry
Files that have been successfully written to the output directory are transferred to this relationship

GatazkaFitxategiaMugitu		
PutFile 2.11.0		
org.apache.nifi - nifi-standard-nar		
In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

7. **Fluxu osoa gatazketan:**

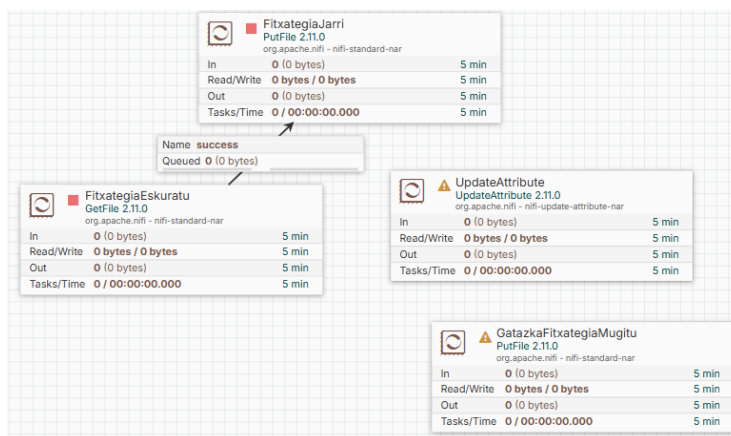
- a. **GetFile -> PutFile #1 (success ohikoa):** Amaiera arrunta.

Arrastatu **GetFile (FitxategiaEskuratu)** prozesadoretik **PutFile (FitxategiaJarri)** prozesadorera.



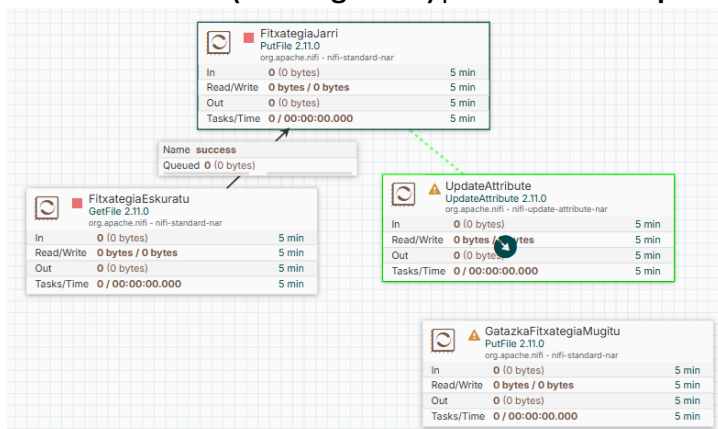
Create Connection

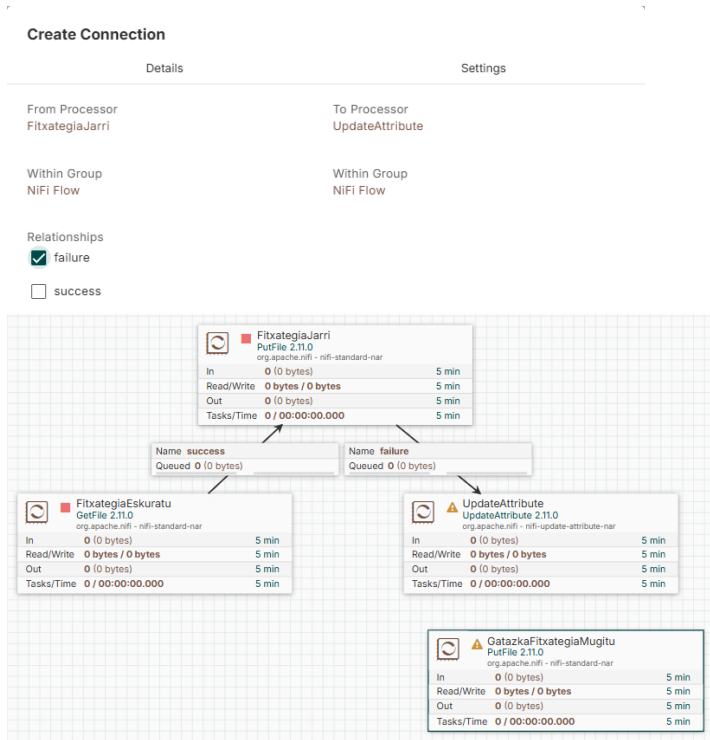
Details	Settings
From Processor FitxategiaEskuratu	To Processor FitxategiaJarri
Within Group NIFI Flow	Within Group NIFI Flow
Relationships ☒ success	



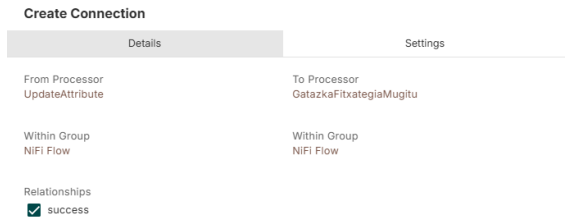
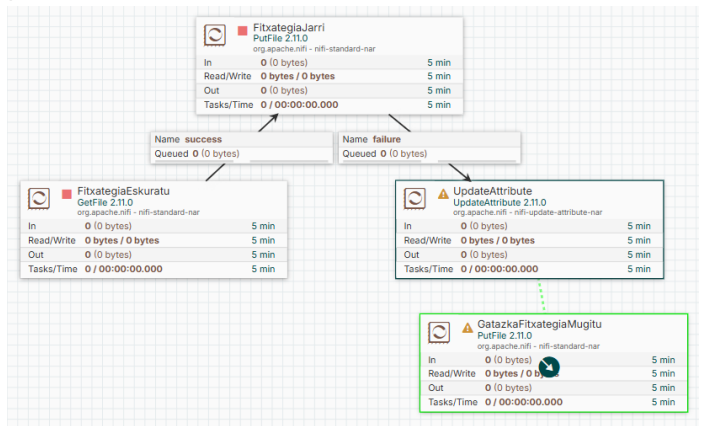
- b. **PutFile #1 (failure) -> UpdateAttribute:** Gatazka kasuetan (izen bereko fitxategia) fitxategiaren izena berritu.

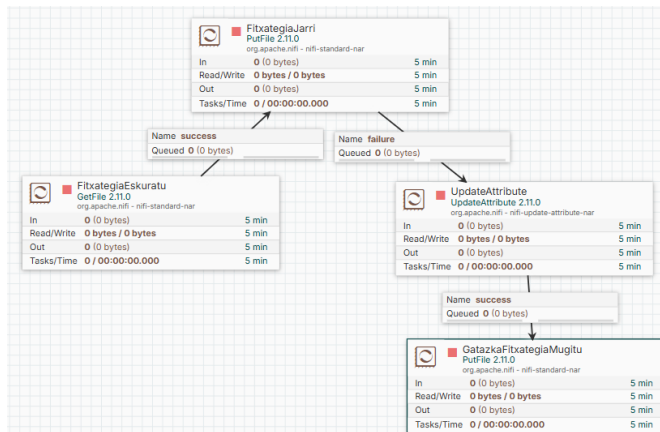
Arrastatu **PutFile (FitxategiaJarri)** prozesadoretik **UpdateAttribute** prozesadorera.





- c. **UpdateAttribute -> PutFile #2 (gatazka-karpeta):** Fitxategiaren izena berriztu.
 Arrastatu **UpdateAttribute** prozesadoretik **PutFile (GatazkaFitxategiaMugitu)** prozesadorera.





Honela, **EZ dago datu-galerarik**; fitxategi guztiak helmugaren batean kokatzen dira.

8. Proba egiteko:

a. Sarrera-karpetan hiru fitxategi sortu (proba_01.txt, proba_02.txt, proba_03.txt):

... ariketak > 01-ariketa-getfile-putfile > sarrera

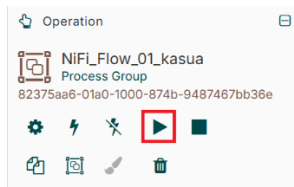
Nombre	Fecha de modificación	Tipo	Tamaño
proba_01	12/09/2026 15:13	Documento de tex...	0 KB
proba_02	12/09/2026 15:13	Documento de tex...	0 KB
proba_03	12/09/2026 15:13	Documento de tex...	0 KB

b. Irteera-karpetan gatazkak karpeta sortu.

... 01-ariketa-getfile-putfile > irteera

Nombre	Fecha de modificación	Tipo	Tamaño
gatazkak	12/09/2026 15:13	Carpeta de archivos	

c. Sortutako fluxua exekutatu:



d. Egiaztatu sarrera-karpetatik kendu eta irteera-karpetara pasatu direla:

... ariketak > 01-ariketa-getfile-putfile > sarrera

Nombre	Fecha de modificación	Tipo	Tamaño
Esta carpeta está vacía.			

... 01-ariketa-getfile-putfile > irteera

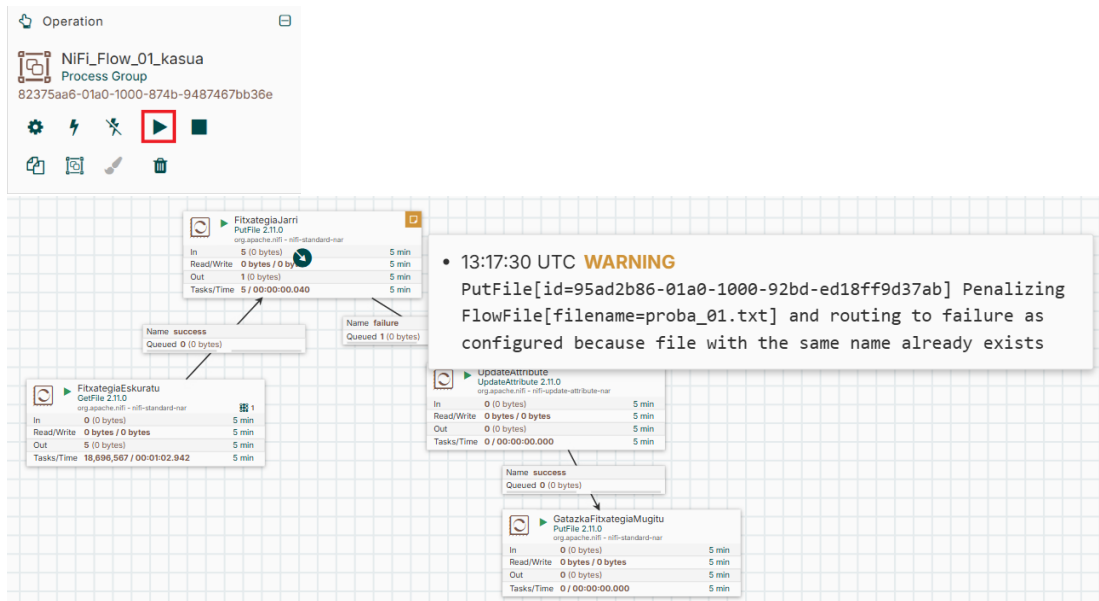
Nombre	Fecha de modificación	Tipo	Tamaño
gatazkak	12/09/2026 15:13	Carpeta de archivos	
proba_01	12/09/2026 15:15	Documento de tex...	0 KB
proba_02	12/09/2026 15:15	Documento de tex...	0 KB
proba_03	12/09/2026 15:15	Documento de tex...	0 KB

e. Sarrera karpetan proba_01.txt izeneko fitxategi bat sortu (gatazka sortuko duena):

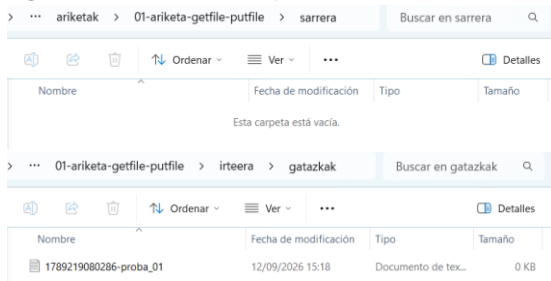
... ariketak > 01-ariketa-getfile-putfile > sarrera

Nombre	Fecha de modificación	Tipo	Tamaño
proba_01	12/09/2026 15:15	Documento de tex...	0 KB

f. Fluxua berriro exekutatu:



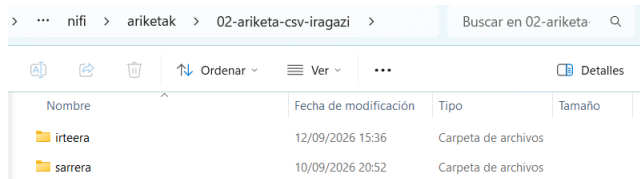
g. Egiaztatu **sarrera** karpetatik kendu eta **irteera/gatazkak** karpetara pasatu direla:



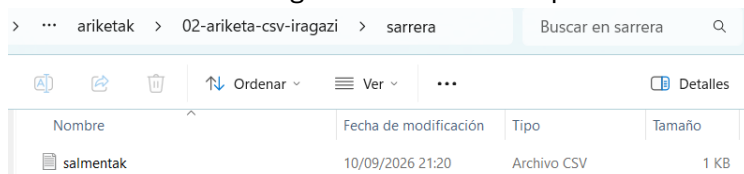
Apache NiFi 2. kasua: CSV datuak iragazi

Helburua: Salmenta-datuak iragazi, **Frantzia** sarrerak eta **Unitateak > 1** soilik gordez.

1. Sarrera-karpeta eta irteera-karpeta sortu Windows-en.



2. **salmentak.csv** fitxategia sortu **sarrera**-karpetan.



```
ProductID;Date;Zip;Units;Revenue;Country
```

```
725;1/15/1999;41540;1;115.5;Germany
```

```
850;2/03/1999;75000;3;245.0;France
```

```
425;3/21/1999;28013;1;87.25;Spain
```

```
725;4/12/1999;75008;5;577.5;France
```

```
910;5/05/1999;10115;2;230.0;Germany
```

```
850;6/30/1999;69001;4;392.0;France
```

```
# Iragazketaren xedea:
```

```
# Country = 'France' ETA Units > 1
```

```
# Helburua: Frantziako salmentak unitate bat baino gehiago
```

```
# duten erregistroak SOILIK pasatzea.
```

```
# Bereizlea: ; (puntu eta koma) - ohikoa Europar formatuan
```

```
# Lehen lerroa: GOIBURUA (zutabe-izenak)
```

3. **Process Group berria:**

ProcessGroup arrastatu, ariketa honi dagozkion elementuak bertan sartzeko. Dagokion izena jarri eta **Add** sakatu.



Create Process Group

Nombre*
NiFiFlow_02_kasua

Parameter Context

Cancel Add

Sortuta dagoenean, klik bikoitza egin **Process Group**-aren gainean, bertan sartzeko.

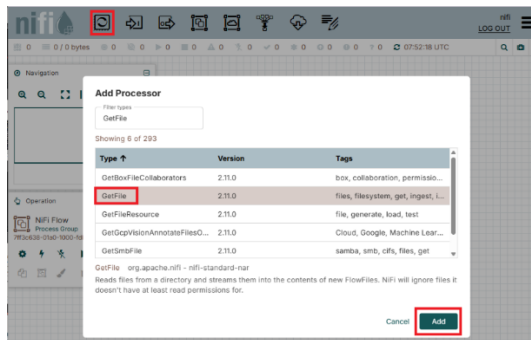
4. **GetFile prozesadorea:**

Gehitu **GetFile** prozesadorea, sarrera-karpetatik fitxategia irakurtzeko:

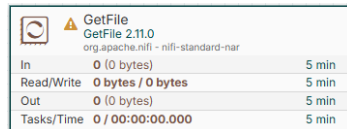
- a. **Processor** arrastatu.



- b. **GetFile** aukeratu eta **Add** sakatu.

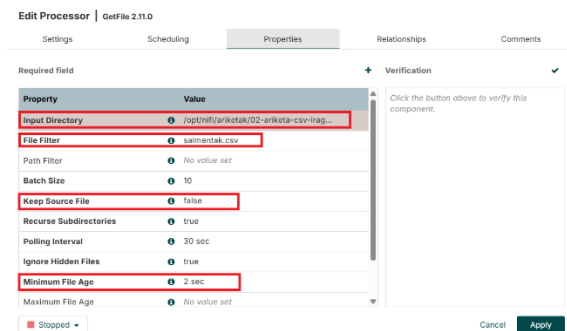


c. Klik bikoitza egin **GetFile**-en gainean, propietateak aldatzeko.



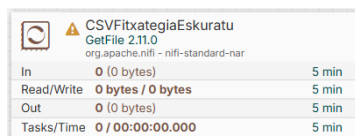
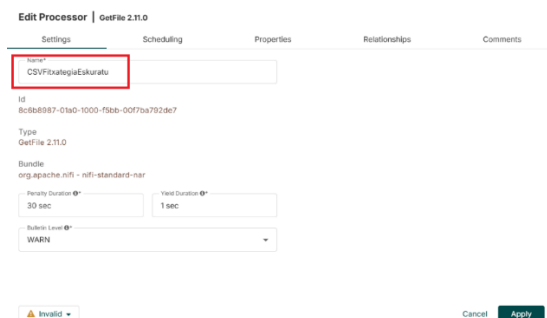
d. **Properties** atalean:

- **Input Directory:** /opt/nifi/ariketak/02-ariketa-csv-iragazi/sarrera
- **Keep Source File:** false
Lehenetsia false da, fitxategiak prozesatu ondoren jatorrizko karpetatik kentzeko.
- **File Filter:** salmentak.csv



e. **Settings** atalean:

- **Name:** CSVFitxategiaEskuratu



5. **SplitRecord** prozesadorea:

Gehitu **SplitRecord** prozesadorea, CSV errenkadak banatzeko:

a. **Processor** arrastatu.



b. **SplitRecord** aukeratu eta **Add** sakatu.

Add Processor

Filter types

splitre

Showing 1 of 293

Type ↑	Version	Tags
SplitRecord	2.11.0	avro, csv, freeform, generic, j...

SplitRecord org.apache.nifi - nifi-standard-nar
Splits up an input FlowFile that is in a record-oriented data format into multiple smaller FlowFiles

Cancel Add

c. Klik bikoitza egin **SplitRecord**-en gainean, propietateak aldatzeko.

	SplitRecord SplitRecord 2.11.0 org.apache.nifi - nifi-standard-nar	
In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

d. **Properties** atalean:

- **Record Reader:** **CSVReader** zerbitzua gehitu *Create new service* aukeraren bidez.

Ondoren, *Go to Service* aukera sakatu eta *Edit* bidez aldatu propietateak:

- **Schema Access Strategy:** **Use String Fields From Header**
- **Value Separator:** ;
- **Treat First Line as Header:** true

Edit Processor | SplitRecord 2.11.0

Settings Scheduling Properties Relationships Comments

Required field Verification ✓

Property	Value
Record Reader	CSVReader
Record Writer	No value set
Records Per Split	No value set

Go to Service Create new service

Click the button above to verify this component.

Controller Services

NIFI Flow → NIFI_Flow_02_kasua

Name ↑	Type	Bundle	State	Scope
CSVReader	CSVReader 2.11.0	org.apache.nifi - nifi-record-serializ...	Disabled	NIFI_Flow_02_kasua

Edit Enable Delete View Documentation

Controller Service Details

Settings Properties Comments

Required field Verification ✓

Property	Value
Schema Access Strategy	Use String Fields From Header
CSV Parser	Apache Commons CSV
Date Format	No value set
Time Format	No value set
Timestamp Format	No value set
CSV Format	Custom Format
Value Separator	;
Record Separator	\n
Treat First Line as Header	true

Close

- **CSVReader** zerbitzua **Enable** gisa jarri.

Controller Services

NIFI Flow → NIFI_Flow_02_kasua

Name ↑	Type	Bundle	State	Scope
CSVReader	CSVReader 2.11.0	org.apache.nifi - nifi-record-serializ...	Disabled	NIFI_Flow_02_kasua
CSVRecordSetWriter	CSVRecordSetWriter 2.11.0	org.apache.nifi - nifi-record-serializ...	Disabled	NIFI_Flow_02_kasua

Edit Enable Delete View Documentation

- **Record Writer: CSVRecordSetWriter** zerbitzua gehitu *Create new service* aukeraren bidez. Ondoren, *Go to Service* aukera sakatu eta *Edit* bidez aldatu propietateak:
 - **Value Separator: ;**

Edit Processor | SplitRecord 2.11.0

Settings Scheduling **Properties** Relationships Comments

Required field + Verification ✓

Property	Value
Record Reader	CSVReader
Record Writer	CSVRecordSetWriter
Records Per Split	1

Click the button above to verify this component.

Go to Service
Create new service

Controller Services

NIFI Flow > NIFI_Flow_01_kassaa_2

Show current scope only

Name	Type	Bundle	State	Scope
CSVReader	CSVReader 2.11.0	org.apache.nifi - nifi-record-serializ...	Enabled	NIFI_Flow_01_kassaa_2
CSVRecordSetWriter	CSVRecordSetWriter 2.11.0	org.apache.nifi - nifi-record-serializ...	Disabled	NIFI_Flow_01_kassaa_2

Edit
Enable
Delete
View Documentation

Edit Controller Service

Settings **Properties** Comments

Required field + Verification ✓

Property	Value
Schema Write Strategy	Do Not Write Schema
Schema Cache	No value set
Schema Access Strategy	Inherit Record Schema
Date Format	No value set
Time Format	No value set
Timestamp Format	No value set
CSV Format	Custom Format
CSV Writer	Apache Commons CSV
Value Separator	;
Include Header Line	true

Click the button above to verify this component.

Cancel Apply

- **CSVRecordSetWriter** zerbitzua **Enable** gisa jarri.

Controller Services

NIFI Flow > NIFI_Flow_02_kassaa

Show current scope only

Name	Type	Bundle	State	Scope
CSVReader	CSVReader 2.11.0	org.apache.nifi - nifi-record-serializ...	Enabled	NIFI_Flow_02_kassaa
CSVRecordSetWriter	CSVRecordSetWriter 2.11.0	org.apache.nifi - nifi-record-serializ...	Disabled	NIFI_Flow_02_kassaa

Edit
Enable
Delete
View Documentation

- **Records per Split: 1** Errenkada bakoitza FlowFile independente, hau da, CSV-ko erregistro bakoitzeko FlowFile berri bat sortuko da ('splits' konexioan ateratzen dira).

Edit Processor | SplitRecord 2.11.0

Settings Scheduling **Properties** Relationships Comments

Required field + Verification ✓

Property	Value
Record Reader	CSVReader
Record Writer	CSVRecordSetWriter
Records Per Split	1

Click the button above to verify this component.

h. **Settings** atalean:

- **Name:** CSVtikFFra

Edit Processor | SplitRecord 2.11.0

Settings Scheduling Properties Relationships Comments

Name * **CSVtkFFra**

Id
95fa9ec4-01a0-1000-f6cd-1dc76b934660

Type
SplitRecord 2.11.0

Bundle
org.apache.nifi - nifi-standard-nar

Penalty Duration ⓘ 30 sec Yield Duration ⓘ 1 sec

Bulletin Level ⓘ WARN

6. **QueryRecord** prozesadorea:

Gehitu **QueryRecord** prozesadorea, SQL bidez iragazketa egiteko:

a. **Processor** arrastatu.



b. **QueryRecord** aukeratu eta **Add** sakatu.

Add Processor

Filter types
queryre

Showing 1 of 293

Type ↑	Version	Tags
QueryRecord	2.11.0	aggregate, avro, calcite, csv, ...

c. Klik bikoitza egin **QueryRecord**-en gainean, propietateak aldatzeko.

QueryRecord
QueryRecord 2.11.0
org.apache.nifi - nifi-standard-nar

In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

d. **Properties** atalean:

- **Record Reader:** CSVReader
- **Record Writer:** CSVRecordSetWriter
- **Include Zero Record FlowFiles:** false
- Propietate pertsonalizatua **FrantziaGehiago1:**
SELECT * FROM FLOWFILE WHERE COUNTRY = 'France' and Units > 1

Edit Processor | QueryRecord 2.11.0

Settings Scheduling **Properties** Relationships Comments

Required field + Verification ✓

Property	Value
Record Reader	CSVReader
Record Writer	CSVRecordSetWriter
Include Zero Record FlowFiles	false
Default Decimal Precision	10
Default Decimal Scale	0
FrantziaGehiago1	SELECT * FROM FLOWFILE WHERE C...

Click the button above to verify this component.

Invalid Cancel Apply

i. **Settings** atalean:

- **Name:** SQLkontsulta

Edit Processor | QueryRecord 2.11.0

Settings Scheduling Properties Relationships Comments

Name*
SQLkontsulta

Id
95fcd28-01a0-1000-cf6d-2c130ea8a401

Type
QueryRecord 2.11.0

Bundle
org.apache.nifi - nifi-standard-nar

Penalty Duration ⓘ*
30 sec

Yield Duration ⓘ*
1 sec

Bulletin Level ⓘ*
WARN

7. UpdateAttribute prozesadorea:

Gehitu **UpdateAttribute** prozesadorea, izen bereko fitxategien gatazka kudeatzeko:

- Processor** arrastatu.



- UpdateAttribute** aukeratu eta **Add** sakatu.

Add Processor

Filter types
updateattr

Showing 1 of 293

Type ↑	Version	Tags
UpdateAttribute	2.11.0	Attribute Expression Language...

- Klik bikoitza egin **UpdateAttribute**-ren gainean, propietateak aldatzeko.

UpdateAttribute
UpdateAttribute 2.11.0
org.apache.nifi - nifi-update-attribute-nar

In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

- Properties** atalean:

- Propietate berria gehitu **+** ikurra sakatuz: **filename**
- Balioa: **`\${filename:substringBeforeLast('.')}\_\${uuid}\_\${now():toNumber()}.csv`**

Edit Processor | UpdateAttribute 2.11.0

Settings Scheduling Properties Relationships Comments

Required field + Verification ✓

Property	Value
Delete Attributes Expression	No value set
Store State	Do not store state
Stateful Variables Initial Value	No value set
Cache Value Lookup Cache Size	100
filename	<code>\${filename:substringBeforeLast('.')}_\${...}</code>

Click the button above to verify this component.

- Settings** atalean:

- Name:** FitxategiaBerrizendatu

Edit Processor | UpdateAttribute 2.11.0

Settings Scheduling Properties Relationships Comments

Name*
FitxategiaBerizendatu

Id
99cf7ba2-01a0-1000-f5ca-46252de4af5b

Type
UpdateAttribute 2.11.0

Bundle
org.apache.nifi - nifi-update-attribute-nar

Penalty Duration ⓘ
30 sec

Yield Duration ⓘ
1 sec

Bulletin Level ⓘ
WARN

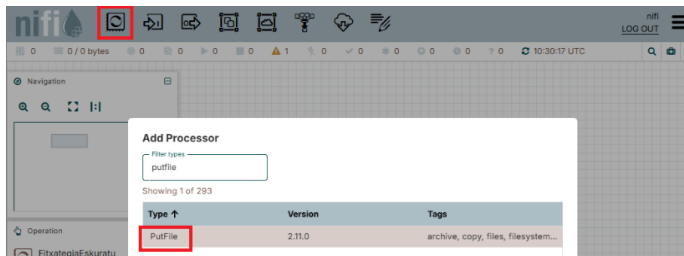
8. **PutFile** prozesadorea:

Gehitu **PutFile** prozesadorea, irteera-karpetan idazteko.

a. **Processor** arrastatu.



b. **PutFile** aukeratu eta **Add** sakatu.



c. Klik bikoitza egin **PutFile**-en gainean, propietateak aldatzeko.

	PutFile PutFile 2.11.0 org.apache.nifi - nifi-standard-nar
In	0 (0 bytes) 5 min
Read/Write	0 bytes / 0 bytes 5 min
Out	0 (0 bytes) 5 min
Tasks/Time	0 / 00:00:00.000 5 min

d. **Properties** atalean:

- **Directory:** /opt/nifi/ariketak/02-ariketa-csv-iragazi/irteera
Helmuga-karpeta absolutua
- **Conflict Resolution Strategy:** fail

Edit Processor | PutFile 2.11.0

Settings Scheduling Properties Relationships Comments

Required field + Verification ✓

Property	Value
Directory	/opt/nifi/ariketak/02-ariketa-csv-iraga...
Conflict Resolution Strategy	fail
Create Missing Directories	true
Maximum File Count	No value set
Last Modified Time	No value set
Permissions	No value set
Owner	No value set
Group	No value set

Click the button above to verify this component.

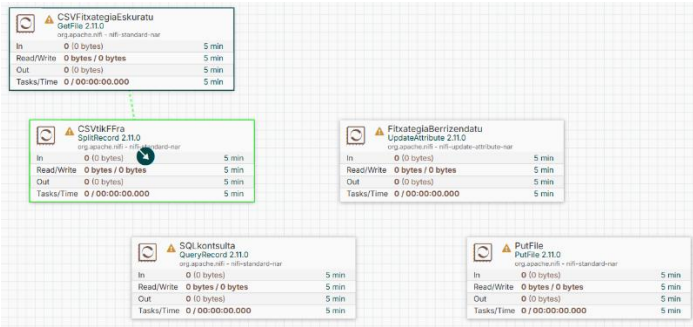
Stopped Cancel Apply

9. **Fluxu osoa. Erlazioak ezarri:**

Sortutako prozesadoreetan erlazioak ezarri (terminate...).

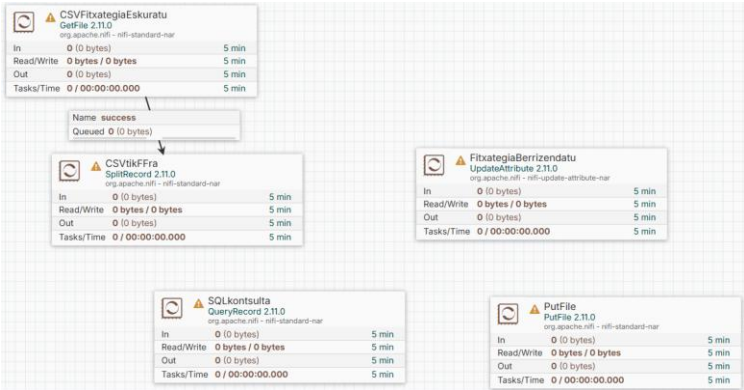
a. **GetFile -> SplitRecord (success ohikoa):** Amaiera arrunta.

Arrastatu **GetFile (CSVFitxategiaEskuratu)** prozesadoretik **SplitRecord (CSVtikFFra)** prozesadorera.

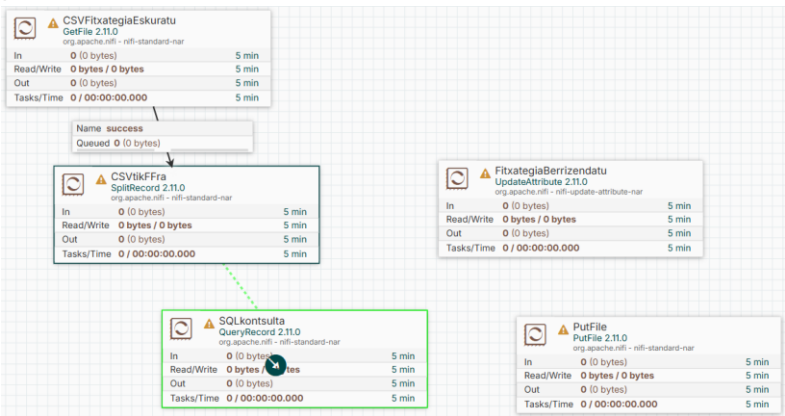


Create Connection

Details	Settings
From Processor CSVFitxategiaEskuratu	To Processor CSVtikFFra
Within Group NIFI_Flow_01_kasua_2	Within Group NIFI_Flow_01_kasua_2
Relationships ☒ success	

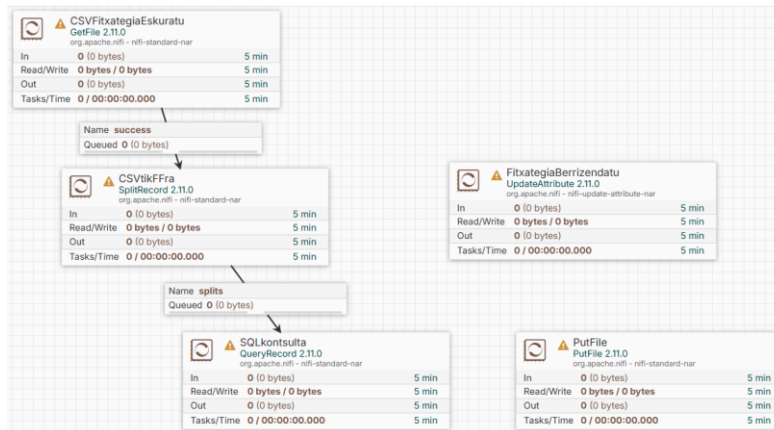


- b. **SplitRecord (failure) -> QueryRecord**: CSVko erregistro bakoitza FlowFile-era pasatuko da, **splits** konexioaren bidez.
- Arrastatu **SplitRecord (CSVtikFFra)** prozesadoretik **QueryRecord (SQLkontsulta)** prozesadorera.



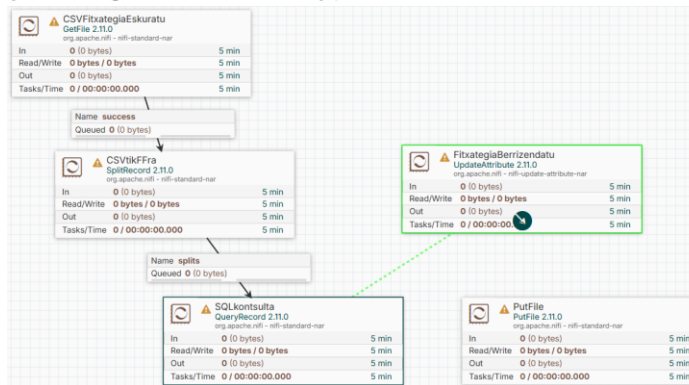
Create Connection

Details	Settings
From Processor CSVtikFFra	To Processor SQLkontsulta
Within Group NIFL_Flow_01_kasua_2	Within Group NIFL_Flow_01_kasua_2
Relationships	
☐ failure ☐ original ☒ splits	



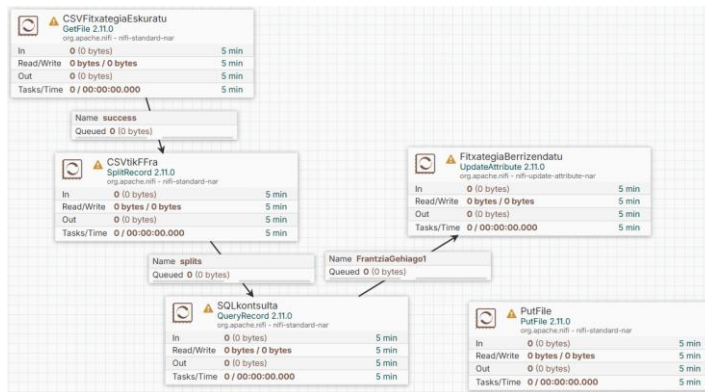
- c. **QueryRecord -> UpdateAttribute:** SQL kontsultaren bidez iragazitako FF-ak pasatuko dira, **FrantziaGehiago1** konexioaren bidez.

Arrastatu **QueryRecord (SQLkontsulta)** prozesadoretik **UpdateAttribute (FitxategiaBerrizendatu)** prozesadorera.



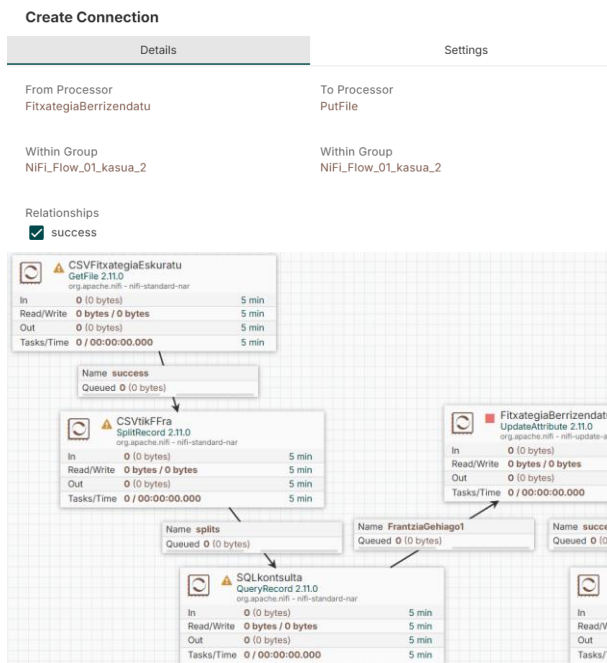
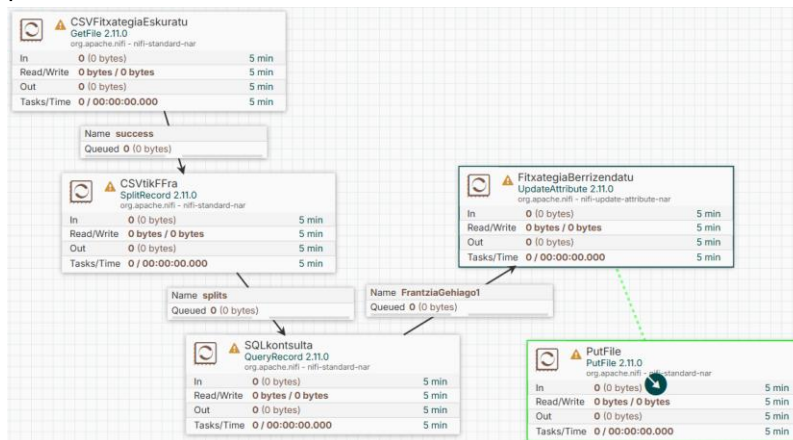
Create Connection

Details	Settings
From Processor SQLkontsulta	To Processor FitxategiaBerrizendatu
Within Group NIFL_Flow_01_kasua_2	Within Group NIFL_Flow_01_kasua_2
Relationships	
☐ failure ☒ FrantziaGehiago1 ☐ original	



d. **UpdateAttribute -> PutFile:** Fitxategiaren izena berriztuta gordeko da.

Arrastatu **UpdateAttribute (FitxategiaBerrizendatu)** prozesadoretik **PutFile** prozesadorera.



e. Konexioak ezarri ondoren, prozesadoreetan lotu gabeko erlazioak **AUTO-TERMINATE** jarri.

- SplitRecord (CSVtikFFra):
 - **Relationships** atalean: **Automatically Terminate/Retry Relationships:**
 - failure: **terminate**

- original: **terminate**

Edit Processor | SplitRecord 2.11.0

Settings	Scheduling	Properties	Relationships	Comments
----------	------------	------------	---------------	----------

Automatically Terminate/Retry Relationships ⓘ

failure

☒ terminate ☐ retry

If a FlowFile cannot be transformed from the configured input format to the configured output format, the unchanged FlowFile will be routed to this relationship.

original

☒ terminate ☐ retry

Upon successfully splitting an input FlowFile, the original FlowFile will be sent to this relationship.

splits

☐ terminate ☐ retry

The individual 'segments' of the original FlowFile will be routed to this relationship.

- QueryRecord (SQL kontsulta):

- Relationships atalean:

Automatically Terminate/Retry Relationships:

- failure: **terminate**
- original: **terminate**

Edit Processor | QueryRecord 2.11.0

Settings	Scheduling	Properties	Relationships	Comments
----------	------------	------------	---------------	----------

Automatically Terminate/Retry Relationships ⓘ

failure

☒ terminate ☐ retry

If a FlowFile fails processing for any reason (for example, the SQL statement contains columns not present in input data), the original FlowFile it will be routed to this relationship

FrantziaGehiagol

☐ terminate ☐ retry

User-defined relationship that specifies where data that matches the specified SQL query should be routed

original

☒ terminate ☐ retry

The original FlowFile is routed to this relationship

- PutFile:

- Relationships atalean:

Automatically Terminate/Retry Relationships:

- failure: **terminate**
- original: **terminate**

Edit Processor | PutFile 2.11.0

Settings	Scheduling	Properties	Relationships	Comments
----------	------------	------------	---------------	----------

Automatically Terminate/Retry Relationships ⓘ

failure

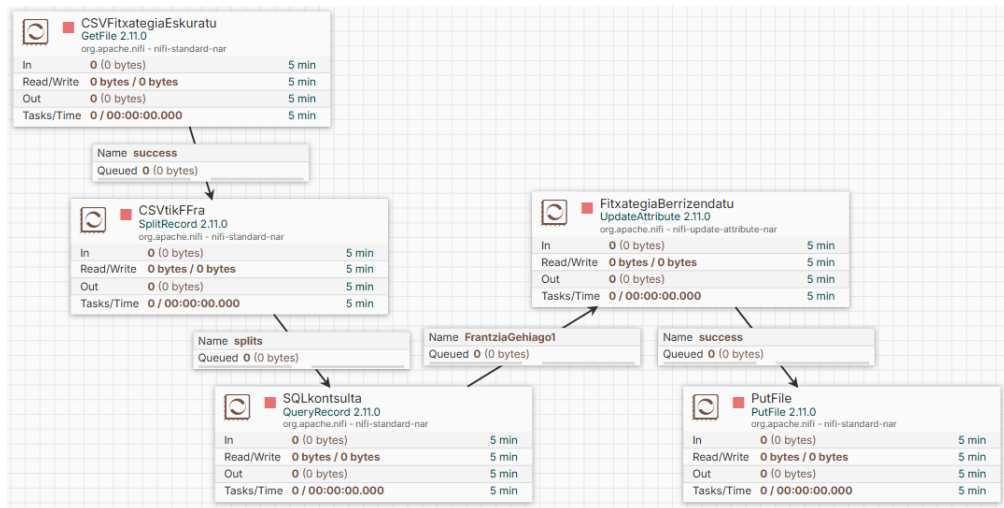
☒ terminate ☐ retry

Files that could not be written to the output directory for some reason are transferred to this relationship

success

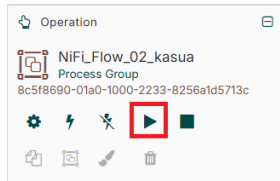
☒ terminate ☐ retry

Files that have been successfully written to the output directory are transferred to this relationship

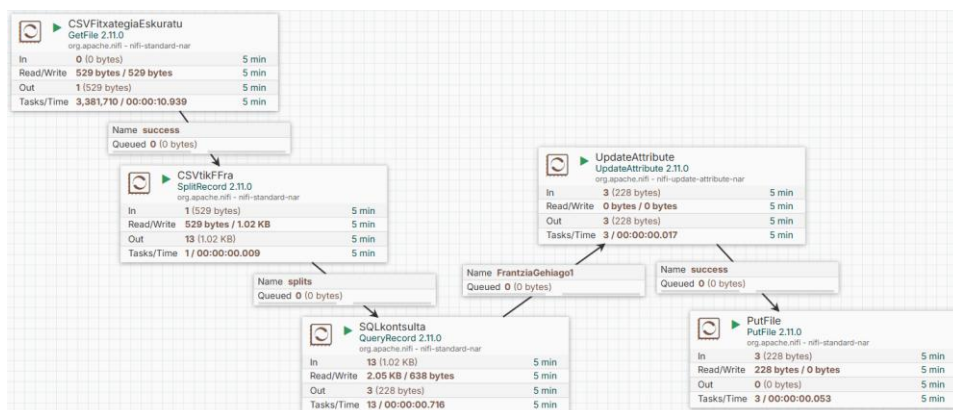
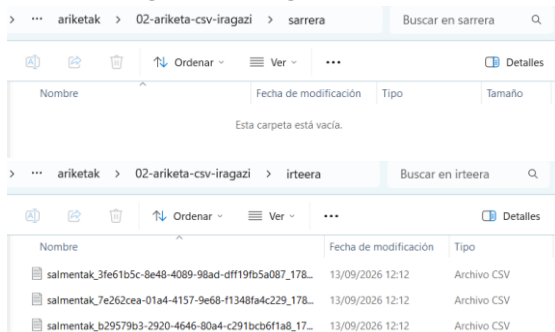


10. Proba egiteko:

a. Sortutako fluxua exekutatu:



b. Egiatzatu **sarrera**-karpetatik **salmentak.csv** kendu eta **irteera**-karpetan berriak sortu direla, erregistroak iragazita.



2. kasuaren 2. aldaera: SplitRecord 10 errenkadako taldeetan

Aldaera: SplitRecord-en **Records per Split** propietatea aldatuko dugu (1-en ordeztu, 10 jarriko dugu). Honela, CSV-ko 10 errenkadak sortuko dira FlowFile-ak.

1. **SplitRecord (CSVtikFFra)** prozesadorean:

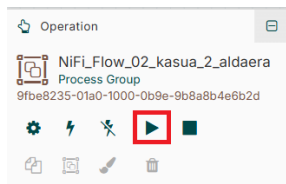
a. **Propietateak** atalean:

- **Records Per Split: 10**

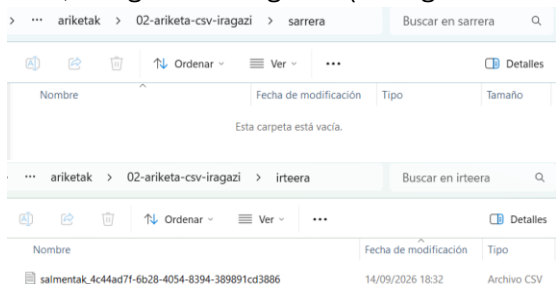
Property	Value
Record Reader	CSVReader
Record Writer	CSVRecordSetWriter
Records Per Split	10

2. Proba egiteko:

a. Sortutako fluxua exekutatu:



b. Egiaztatu **sarrera**-karpetatik **salmentak.csv** kendu eta **irteera**-karpetan berria sortu dela, erregistroak iragazita (oraingoan fitxategi bakarrean hiru erregistro).

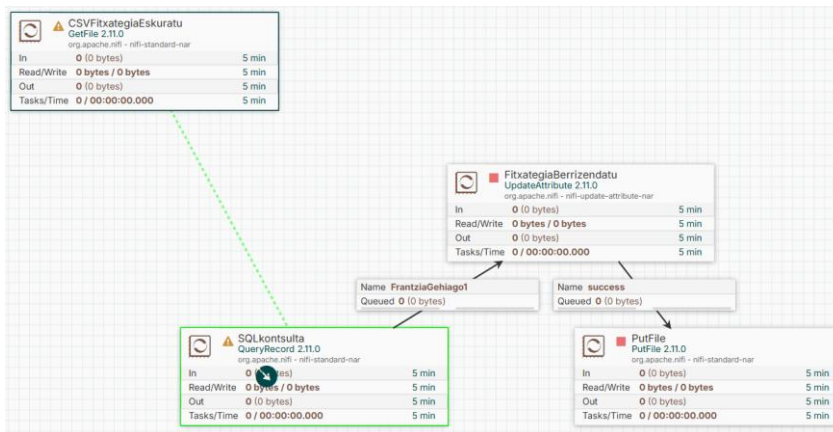


2. kasuaren 3. aldaera (optimizazioa): SplitRecord gabe

Arazoa: SplitRecord-ek FlowFile asko sortzen ditu (CSV-ko errenkada bakoitzeko bat). Honek memoria-zama handia eragin dezake.

Soluzioa: GetFile zuzenean QueryRecord-era konektatu, SQL fitxategi osoaren gainean exekuta dadin. Honela, QueryRecord-ek barnetik aplikatzen du iragazketa Calcite motorrarekin, banaketa beharrik gabe. FlowFile kopurua murriztuko da, errendimendua hobetuko da eta memoria-erabilera txikiagoa izango da.

1. **GetFile (CSVFitxategiaEskuratu)** prozesadoretik **SplitRecord (CSVtikFFra)** doan konexioa ezabatu.
2. **SplitRecord (CSVtikFFra)** prozesadorea ezabatu.
3. **GetFile (CSVFitxategiaEskuratu)** prozesadoretik **QueryRecord (SQLkontsulta)** prozesadorera konexioa ezarri.

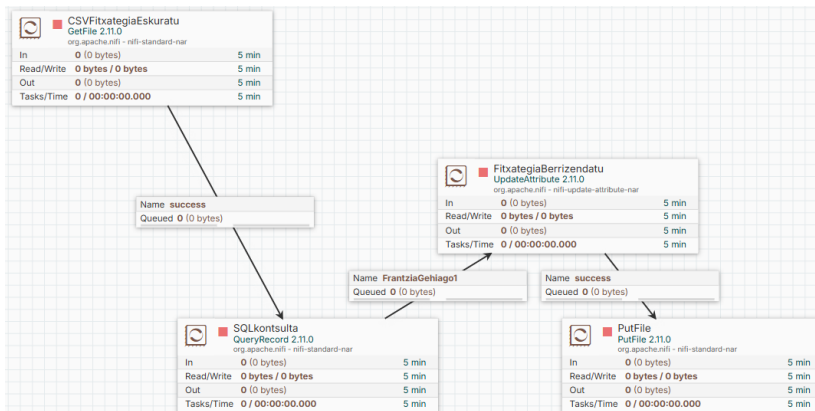


Create Connection

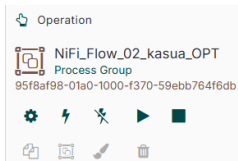
Details	Settings
From Processor CSVFitxategiaEskuratu	To Processor SQLkotsulta
Within Group NiFi_Flow_02_kasua_OPT	Within Group NiFi_Flow_02_kasua_OPT

Relationships

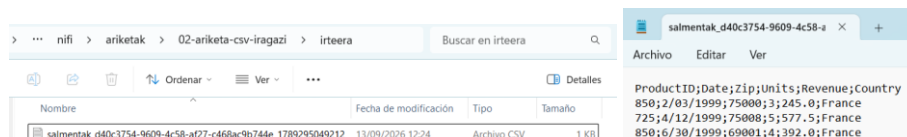
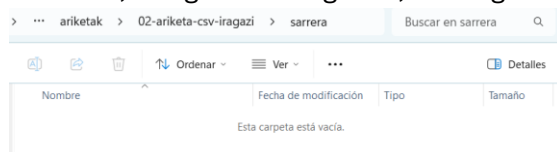
☒ success



4. **GetFile (CSVFitxategiaEskuratu)** prozesadoretik **QueryRecord (SQLkotsulta)** prozesadorera konexioa ezarri.
5. Fluxua exekutatu probatzeko:



6. Egiaztatu **sarrera**-karpetatik **salmentak.csv** kendu eta **irteera**-karpetan fitxategi berria sortu dela, erregistroak iragazita, fitxategi bakar batean.



Apache NiFi 3. kasua: Atributuak eta Linajea

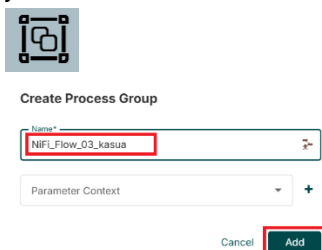
Helburua: FlowFile baten **edukia** eta **atributuak** bereiztea, edukitik atributu bat erauztea, atributuak logean erregistratzea, edukia fitxategi batean gordetzea eta FlowFile-aren datu-linajea aztertzea.

Erabiliko ditugun prozesadoreak:

- GenerateFlowFile (testing)
- ReplaceText
- ExtractText
- LogAttribute
- PutFile

1. **Process Group berria:**

ProcessGroup arrastatu, ariketa honi dagozkion elementuak bertan sartzeko. Dagokion izena jarri eta **Add** sakatu.



Sortuta dagoenean, klik bikoitza egin **Process Group**-aren gainean, bertan sartzeko.

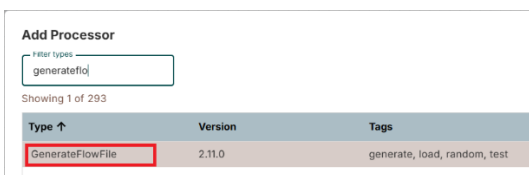
2. **GenerateFlowFile prozesadorea:**

Gehitu **GenerateFlowFile** prozesadorea, FF sortzeko:

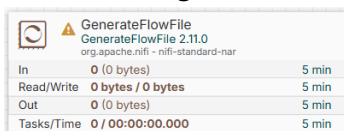
a. **Processor** arrastatu.



b. **GenerateFlowFile** aukeratu eta **Add** sakatu.



c. Klik bikoitza egin **GenerateFlowFile** -en gainean, propietateak aldatzeko.



d. **Properties** atalean:

- **File Size: 10B**
Sortzen den edukiaren tamaina.
- **Batch Size: 1**
Sortzen den FlowFile kopurua exekuzio bakoitzeko.
- **Data Format: Text**
- **Unique FlowFiles: true** (FF bakoitza ezberdina; UUID-unikoa eduki gisa)

Edit Processor | GenerateFlowFile 2.11.0

Settings Scheduling **Properties** Relationships Comments

Required field + Verification ✓

Property	Value
File Size	10B
Batch Size	1
Data Format	Text
Unique FlowFiles	true
Custom Text	No value set
Character Set	UTF-8
Mime Type	No value set

Click the button above to verify this component.

e. **Scheduling** atalean:

- **Run Schedule: 3s** (segunduro berriz exekutatu)

Edit Processor | GenerateFlowFile 2.11.0

Settings **Scheduling** Properties Relationships Comments

Scheduling Strategy: Timer driven

Concurrent Tasks: 1

Run Schedule: 3 min

Run Duration: [Slider]

Execution: All nodes

Lower latency Higher throughput

3. **ReplaceText** prozesadorea:

Gehitu **ReplaceText** prozesadorea, FlowFile-aren edukia aldatzeko:

a. **Processor** arrastatu.



b. **ReplaceText** aukeratu eta **Add** sakatu.

Add Processor

Filter types: replacertext

Showing 2 of 293

Type ↑	Version	Tags
ReplaceText	2.11.0	Change, Modify, Regex, Regul...
ReplaceTextWithMapping	2.11.0	Change, Mapping, Modify, Re...

c. Klik bikoitza egin **ReplaceText**-en gainean, propietateak aldatzeko.

ReplaceText
ReplaceText 2.11.0
org.apache.nifi - nifi-standard-nar

In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

d. **Properties** atalean:

- **Replacement Strategy: Always Replace**

Aukerak:

- **Always Replace:** Edukia OSORIK berridatzi
- **Regex Replace:** Regex-arabera ordezkatu
- **Literal Replace:** Testu konkretua beste batekin ordeztu (regex-rik gabe)
- **Append/Prepend:** Eduki existenteari testua ondoren/aurretik gehitu

- **Replacement Value: proba** ('proba' adibidez, edo Expression Language erabiliz: `${now()}`)

Aukerak:

- **Testua** (adibidez, 'proba')
- **Expression Language** erabil daiteke, atributuetatik balioak hartzeko (`${filename}`, `${uuid}`, `${now()}`...)

Edit Processor | ReplaceText 2.11.0

Settings Scheduling **Properties** Relationships Comments

Required field + Verification ☒

Property	Value
Replacement Strategy	Always Replace
Replacement Value	proba
Character Set	UTF-8
Maximum Buffer Size	1 MB
Evaluation Mode	Line-by-Line
Line-by-Line Evaluation Mode	All

Click the button above to verify this component.

e. **Relationships** atalean:

- **Automatically Terminate/Retry Relationships:**
 - failure: **terminate** (konektatu ez dugun harremana ezabatu)

Edit Processor | ReplaceText 2.11.0

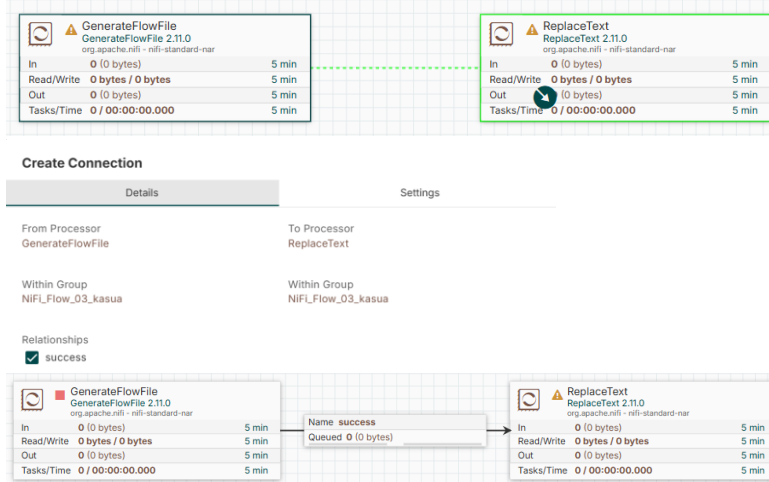
Settings Scheduling Properties **Relationships** Comments

Automatically Terminate/Retry Relationships ☒

failure
☒ terminate ☐ retry
 FlowFiles that could not be updated are routed to this relationship

success
☐ terminate ☐ retry
 FlowFiles that have been successfully processed are routed to this relationship. This includes both FlowFiles that had text replaced and those that did not.

4. **GenerateFlowFile prozesadorea ReplaceText prozesadorearekin konektatu:**



5. **ExtractText prozesadorea**

Gehitu **ExtractText** prozesadorea, edukia atributura bihurtzeko.

Oso prozesadore baliagarria da: askotan beharrezkoa da edukia 'ulertzea' eta atributuetan banatzea, gerora bideraketa edo agregaziorako.

Prozesadorea konfiguratu, FF-aren edukia hartu eta atributu berri batean ipini dezan.

a. **Processor** arrastatu.



b. **ExtractText** aukeratu eta **Add** sakatu.

Add Processor

Filter types: extracttext

Showing 1 of 293

Type ↑	Version	Tags
ExtractText	2.11.0	evaluate, extract, regex, Regul...

c. Klik bikoitza egin **ExtractText**-en gainean, propietateak aldatzeko.

ExtractText ExtractText 2.11.0 org.apache.nifi - nifi-standard-nar			
In	0 (0 bytes)		5 min
Read/Write	0 bytes / 0 bytes		5 min
Out	0 (0 bytes)		5 min
Tasks/Time	0 / 00:00:00.000		5 min

d. **Properties** atalean:

- Propietate berria gehitu **+** ikurra sakatuz: **edukia**
- Regex balioa: **(.*)**
- **Include Capture Group 0: false**

Konexioa **matched** izango da.

Edit Processor | ExtractText 2.11.0

Settings | Scheduling | **Properties** | Relationships | Comments

Required field

Property	Value
Character Set	UTF-8
Maximum Buffer Size	1 MB
Maximum Capture Group Length	1024
Enable Canonical Equivalence	false
Enable Case-Insensitive Matching	false
Permit Whitespace and Comment...	false
Include Capture Group 0	false
Enable Repeating Capture Group	false
Enable Named Groups	false
edukia	(.*)

Invalid

Cancel Apply

e. **Relationships** atalean:

- **Automatically Terminate/Retry Relationships:**
 - unmatched: **terminate** (konektatu ez dugun harremana ezabatu)

Edit Processor | ExtractText 2.11.0

Settings | Scheduling | Properties | **Relationships** | Comments

Automatically Terminate/Retry Relationships ⓘ

matched

☐ terminate ☐ retry

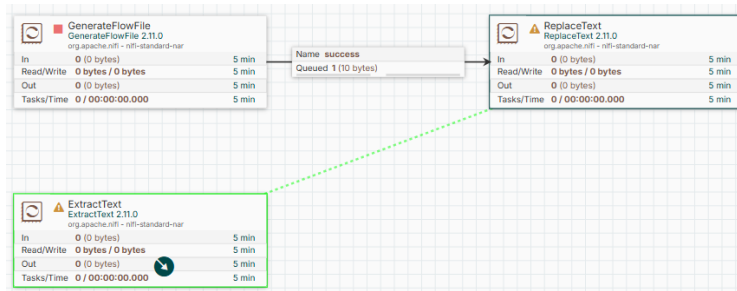
FlowFiles are routed to this relationship when the Regular Expression is successfully evaluated and the FlowFile is modified as a result

unmatched

☒ terminate ☐ retry

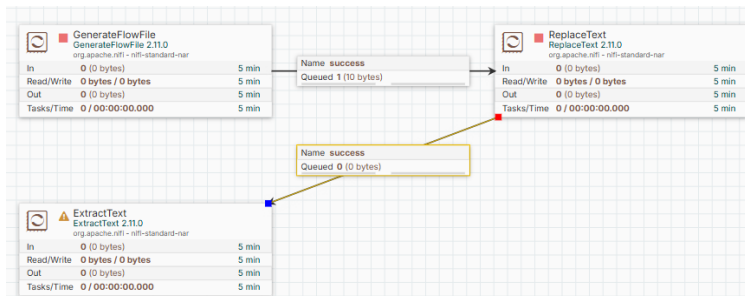
FlowFiles are routed to this relationship when no provided Regular Expression matches the content of the FlowFile

6. **ReplaceText prozesadorea ExtractText prozesadorearekin konektatu:**



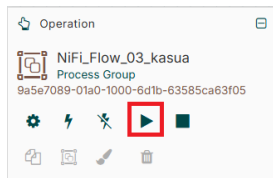
Create Connection

Details	Settings
From Processor ReplaceText	To Processor ExtractText
Within Group NiFi_Flow_03_kasua	Within Group NiFi_Flow_03_kasua
Relationships	
☐ failure ☒ success	

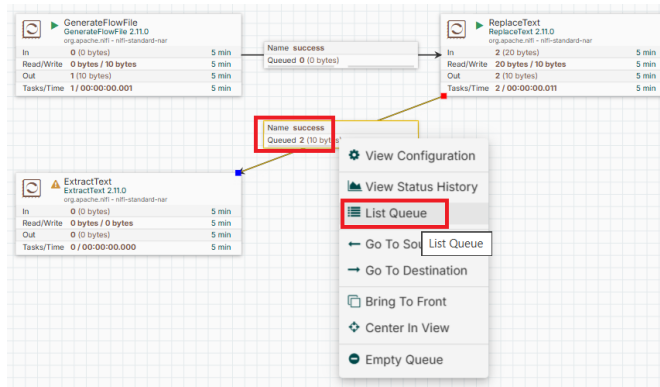


7. Ilara ikusteko:

a. Fluxua exekutatu:



b. Ilaran eskuineko botoiarekin klik egin eta List Queue aukeratu:



success

Display 2 of 2 (10.00 bytes) bytes The source and destination of this queue are currently running. This listing may no longer be accurate.

Position	UUID	Filename	File ...	Que...	Lin...	Penalized
1	3cad7232-858d-4afd-a54...	3cad7232-858d-4afd-a54...	5.00...	00:...	01:2...	No
2	efe125fc-f524-4027-9a0b...	efe125fc-f524-4027-9a0b...	5.00...	00:...	00:...	

View details
Download content
View content
Provenance

View
text
mime.type unknown

Formatted Original

1 proba

8. **LogAttribute** prozesadorea

Gehitu **LogAttribute** prozesadorea, pipeline-aren egoera monitorizatzeko (FF-aren atributuak NiFi-aren log-fitxategian idazten ditu: **nifi-app.log**).

NiFi-aren log egitura:

- nifi-app.log: pipeline-aren log nagusia
- nifi-bootstrap.log: NiFi-aren abioa
- nifi-user.log: autentifikazio eta UI ekintzak

LogAttribute debugging-erako gakoa da. Oso lagungarria da produkzio-pipelinetan arazoak diagnostikatzeko (FF bakoitzaren atributu eta egoera erregistratzeko).

a. **Processor** arrastatu.



b. **LogAttribute** aukeratu eta **Add** sakatu.

Add Processor

Filter types
logat

Showing 1 of 293

Type ↑	Version	Tags
LogAttribute	2.11.0	attributes, logging

c. Klik bikoitza egin **LogAttribute**-en gainean, propietateak aldatzeko.

LogAttribute
LogAttribute 2.11.0
org.apache.nifi - nifi-standard-nar

In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

d. **Properties** atalean:

- Log Level: **info**

Aukerak:

- **info** (ohikoa, gomendatua)
- **iebug** (denbora errealean bideratze-arazoak ikertzeko)
- **warn / error** (egiazko arazoetarako)

Edit Processor | LogAttribute 2.11.0

Settings | Scheduling | Properties | Relationships | Comments

Required field + Verification ✓

Property	Value
Log Level	Info
Log Payload	false
Attributes to Log	No value set
Attributes to Log Regular Expression	.*
Attributes to Ignore	No value set
Attributes to Ignore Regular Expression	No value set
Log FlowFile Properties	true
Output Format	Line per Attribute
Log Prefix	No value set
Character Set	UTF-8

Click the button above to verify this component.

Invalid Cancel Apply

e. Relationships atalean:

- Automatically Terminate/Retry Relationships:
 - success: terminate (konektatu ez dugun harremana ezabatu)

Edit Processor | LogAttribute 2.11.0

Settings | Scheduling | Properties | Relationships | Comments

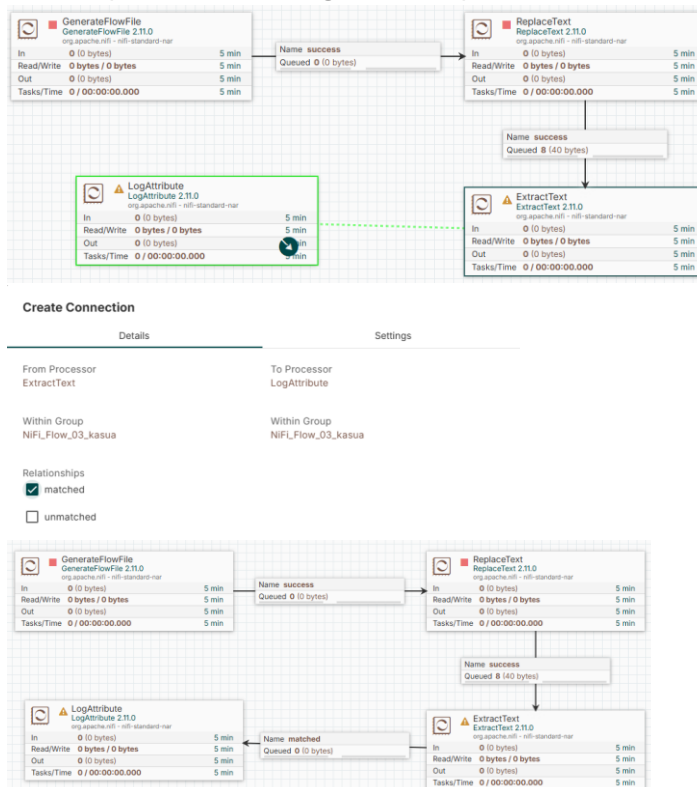
Automatically Terminate/Retry Relationships ⓘ

SUCCESS

☒ terminate ☐ retry

All FlowFiles are routed to this relationship

9. ExtractText prozesadorea LogAttribute prozesadorearekin konektatu:



10. PutFile prozesadorea

Gehitu PutFile prozesadorea, informazioa gordetzeko.

a. Processor arrastatu.



b. PutFile aukeratu eta Add sakatu.

Add Processor

Filter types

putFile

Showing 1 of 293

Type ↑	Version	Tags
PutFile	2.11.0	archive, copy, files, filesystem...

f. Klik bikoitza egin **PutFile**-en gainean, propietateak aldatzeko.

PutFile 2.11.0 org.apache.nifi - nifi-standard-nar		
In	0 (0 bytes)	5 min
Read/Write	0 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	0 / 00:00:00.000	5 min

g. **Properties** atalean:

- **Directory:** /opt/nifi/ariketak/03-ariketa-atributuak-linajea/irteera
- **Conflict Resolution Strategy:** fail

Edit Processor | PutFile 2.11.0

Settings Scheduling **Properties** Relationships Comments

Required field + Verification ✓

Property	Value
Directory	/opt/nifi/ariketak/03-ariketa-atributuak...
Conflict Resolution Strategy	fail
Create Missing Directories	true
Maximum File Count	No value set
Last Modified Time	No value set
Permissions	No value set
Owner	No value set
Group	No value set

Click the button above to verify this component.

Stopped Cancel Apply

h. **Relationships** atalean:

- **Automatically Terminate/Retry Relationships:**
 - failure: terminate
 - success: terminate

Edit Processor | PutFile 2.11.0

Settings Scheduling Properties **Relationships** Comments

Automatically Terminate/Retry Relationships

failure

☒ terminate ☐ retry

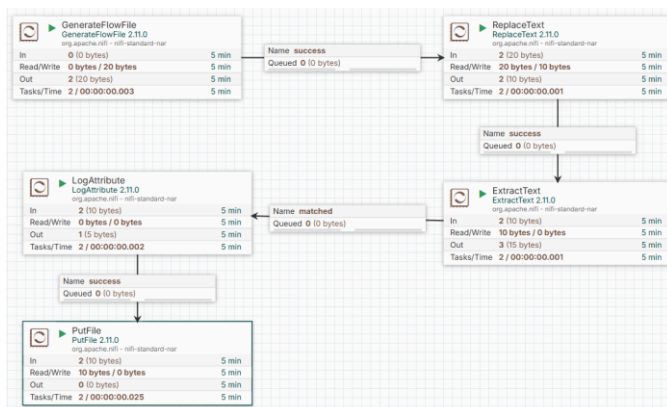
Files that could not be written to the output directory for some reason are transferred to this relationship

success

☒ terminate ☐ retry

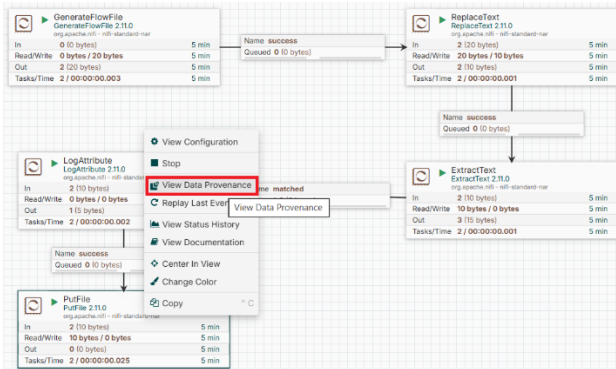
Files that have been successfully written to the output directory are transferred to this relationship

11. **ExtractText** prozesadorea **PutFile** prozesadorearekin konektatu:



12. Datuen linajea ikusi:

- a. Prozesadore baten gainean **eskuineko botoiarekin** klik egin -> **'View data provenance'**



- b. Denbora-zerrenda agertzen da prozesatze-saio guztiekin.

Provenance

Oldest event available: 09/08/2026 18:20:13 UTC

Showing the events that match the specified query. [Clear Sel](#)

Filter: component name

Filter matched 41 of 41

Event Time	Type	FlowFile UUID	File Size	Component Name	Component Type
09/14/2026 19:11:37.496 UTC	DROP	ea05a78d-bd29-48fb-9b18-4b644ed4...	5 bytes	PutFile	PutFile
09/14/2026 19:11:37.495 UTC	SEND	ea05a78d-bd29-48fb-9b18-4b644ed4...	5 bytes	PutFile	PutFile
09/14/2026 19:08:37.512 UTC	DROP	195ded77-0449-4c46-8818-9783b1355...	5 bytes	PutFile	PutFile
09/14/2026 19:08:37.512 UTC	SEND	195ded77-0449-4c46-8818-9783b1355...	5 bytes	PutFile	PutFile
09/14/2026 19:07:14.448 UTC	DROP	65e95236-122b-4121-9e53-de4fc21dd...	5 bytes	PutFile	PutFile
09/14/2026 19:07:14.448 UTC	SEND	65e95236-122b-4121-9e53-de4fc21dd...	5 bytes	PutFile	PutFile

- c. Saio bat aukeratu:

- View Details:

Provenance

Oldest event available: 09/08/2026 18:20:13 UTC

Showing the events that match the specified query. [Clear Sel](#)

Filter: component name

Filter matched 41 of 41

Event Time	Type	FlowFile UUID	File Size	Component Name	Component Type
09/14/2026 19:11:37.496 UTC	DROP	ea05a78d-bd29-48fb-9b18-4b644ed4...	5 bytes	PutFile	PutFile
09/14/2026 19:11:37.495 UTC	SEND	ea05a78d-bd29-48fb-9b18-4b644ed4...	5 bytes	PutFile	PutFile
09/14/2026 19:08:37.512 UTC	DROP	195ded77-0449-4c46-8818-9783b1355...	5 bytes	PutFile	PutFile
09/14/2026 19:08:37.512 UTC	SEND	195ded77-0449-4c46-8818-9783b1355...	5 bytes	PutFile	PutFile

Provenance Event

Details

Time: 09/14/2026 19:11:37.496 UTC

Event Duration: 00:00:00.012

Lineage Duration: 00:00:00.047

Type: DROP

FlowFile UUID: ea05a78d-bd29-48fb-9b18-4b644ed47db6

File Size: 5 bytes

Component Id: 9b029288-01a0-1000-9a70-67d9c0f0860d

Component Name: PutFile

Component Type: PutFile

Details

Attributes

Parent FlowFiles (0)
No parents

Child FlowFiles (0)
No children

Content

Ok

- Show Lineage:

Provenance

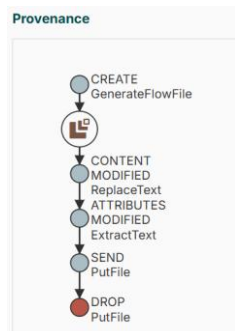
Oldest event available: 09/08/2026 18:20:13 UTC

Showing the events that match the specified query. [Clear Sel](#)

Filter: component name

Filter matched 41 of 41

Event Time	Type	FlowFile UUID	File Size	Component Name	Component Type
09/14/2026 19:11:37.496 UTC	DROP	ea05a78d-bd29-48fb-9b18-4b644ed4...	5 bytes	PutFile	PutFile
09/14/2026 19:11:37.495 UTC	SEND	ea05a78d-bd29-48fb-9b18-4b644ed4...	5 bytes	PutFile	PutFile
09/14/2026 19:08:37.512 UTC	DROP	195ded77-0449-4c46-8818-9783b1355...	5 bytes	PutFile	PutFile
09/14/2026 19:08:37.512 UTC	SEND	195ded77-0449-4c46-8818-9783b1355...	5 bytes	PutFile	PutFile
09/14/2026 19:07:14.448 UTC	DROP	65e95236-122b-4121-9e53-de4fc21dd...	5 bytes	PutFile	PutFile



Grafiko bat agertzen da FF horren bidaia osoa erakutsiz, prozesadore bakoitzak zer eraldaketa egin duen (atributu berriak, edukia aldatuta...)

13. Log fitxategian gordetzen duena ikusteko (nifi-app.log):

- Powershell-en, log-a erakusteko komandoa exekutatu:
docker exec -it nifi tail -f /opt/nifi/nifi-current/logs/nifi-app.log
- Fluxua exekutatu:

```

2026-09-14 18:43:14,513 INFO [Timer-Driven Process Thread-8] o.a.n.processors.standard.LogAttribute
LogAttribute[id=9ae58493-01a0-1000-6f7b-334ff34f0743] logging for FlowFile StandardFlowFileRecor
d[uuid=86c2560e-6653-4d54-8ab2-facf6876bce7,claim=StandardContentClaim [resourceClaim=StandardReso
urceClaim[id=1789403560360-1, container=default, section=1], offset=1323, length=5],offset=0,name=
1f313533-4418-4585-a5f5-d14a0ccfd137,size=5]
-----
FlowFile Properties
Key: 'entryDate'
  Value: 'Mon Sep 14 18:43:14 UTC 2026'
Key: 'lineageStartDate'
  Value: 'Mon Sep 14 18:43:14 UTC 2026'
Key: 'fileSize'
  Value: '5'
FlowFile Attribute Map Content
Key: 'edukia'
  Value: 'proba'
Key: 'filename'
  Value: '1f313533-4418-4585-a5f5-d14a0ccfd137'
Key: 'path'
  Value: './'
Key: 'uuid'
  Value: '86c2560e-6653-4d54-8ab2-facf6876bce7'
-----
2026-09-14 18:43:14,710 INFO [Flow Service Tasks Thread-1] o.a.nifi.controller.StandardFlowService
Saved flow controller org.apache.nifi.controller.FlowController@58f2605b // Another save pending
= false
2026-09-14 18:43:33,713 INFO [Checkpoint FlowFile Repository] o.a.n.wali.SequentialAccessWriteAhea
dLog Checkpointed Write-Ahead Log with 0 Records and 0 Swap Files in 4 milliseconds (Stop-the-world
d time = 2 milliseconds), max Transaction ID 12632
2026-09-14 18:43:33,714 INFO [Checkpoint FlowFile Repository] o.a.n.c.r.WriteAheadFlowFileReposito
ry Successfully checkpointed FlowFile Repository with 0 records in 4 milliseconds
  
```

Produktzio-erabilera:

- Auditoretza. Datu sentikorrekina lan egiten denean (osasuna, finantza...), presio legal handia jarraipenari buruz.
- Debugging-erako. FlowFile konkretu batek arazoak baditu, bere bidaia osoa azter daiteke arazoaren jatorria identifikatzeko.

3. kasuaren 2. aldaera:

Egoera: GenerateFlowFile prozesadoreak FlowFile bat sortuko du segunduro. ReplaceText-ek aurreko edukia mantenduko du eta `[[[ Data: ${now()} ]]]` testua gehituko dio (Append). Ondoren, ExtractText erabiliz atributuak logean erregistratuko ditu. Azkenik, AttributesToJSON erabiliz *datuak* atributua JSON dokumentu bihurtuko du eta PutMongo erabiliz MongoDB datu-basean gordeko du.

Helburua: pipeline pertsonalizatua + MongoDB integrazioa.

1. GenerateFlowFile prozesadorea:

Edit Processor | GenerateFlowFile 2.11.0

Settings | **Scheduling** | Properties | Relationships | Comments

Scheduling Strategy ⓘ* Timer driven
The strategy used to schedule this processor.

Concurrent Tasks ⓘ* 1

Run Schedule ⓘ* 5 sec

Run Duration ⓘ* Lower latency Higher throughput

Execution ⓘ* All nodes

2. ReplaceText prozesadorea:

Edit Processor | ReplaceText 2.11.0

Settings | Scheduling | **Properties** | Relationships | Comments

Required field + Verification ✓

Property	Value
Replacement Strategy ⓘ	Append
Replacement Value ⓘ	'[[[Data: \${now()}]]]'
Character Set ⓘ	UTF-8
Maximum Buffer Size ⓘ	1 MB
Evaluation Mode ⓘ	Line-by-Line
Line-by-Line Evaluation Mode ⓘ	All

Click the button above to verify this component.

3. ExtractText prozesadorea:

Edit Processor | ExtractText 2.11.0

Settings | Scheduling | **Properties** | Relationships | Comments

Required field + Verification ✓

Property	Value
Permit Whitespace and Comment... ⓘ	false
Enable DOTALL Mode ⓘ	false
Enable Literal Parsing of the Pattern ⓘ	false
Enable Multiline Mode ⓘ	false
Enable Unicode-aware Case Folding ⓘ	false
Enable Unicode Predefined Chara... ⓘ	false
Enable Unix Lines Mode ⓘ	false
Include Capture Group 0 ⓘ	false
Enable Repeating Capture Group ⓘ	false
Enable Named Groups ⓘ	false
datuak ⓘ	(.*)

Click the button above to verify this component.

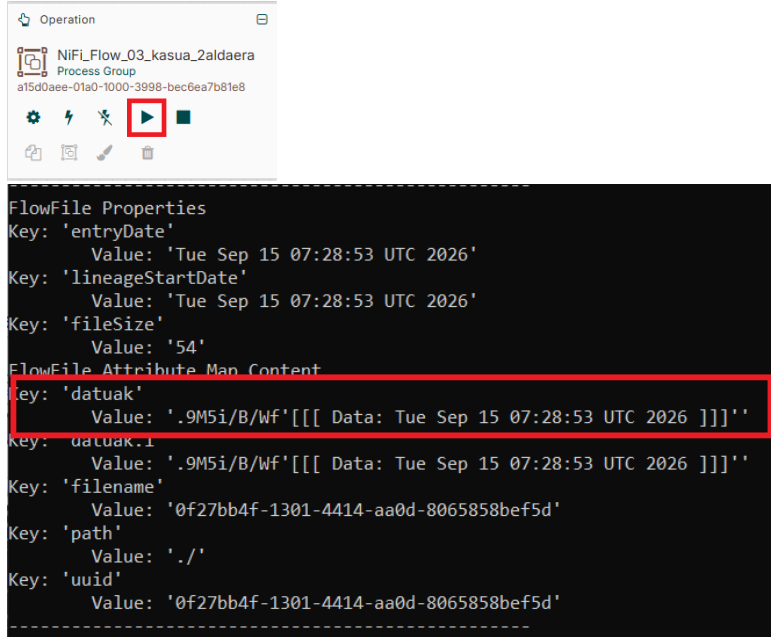
Stopped Cancel Apply

4. LogAttribute prozesadorea:

- a. Prozesadoreak ez du behar aldaketarik. Log-a probatzeko:

docker exec -it nifi tail -f /opt/nifi/nifi-current/logs/nifi-app.log

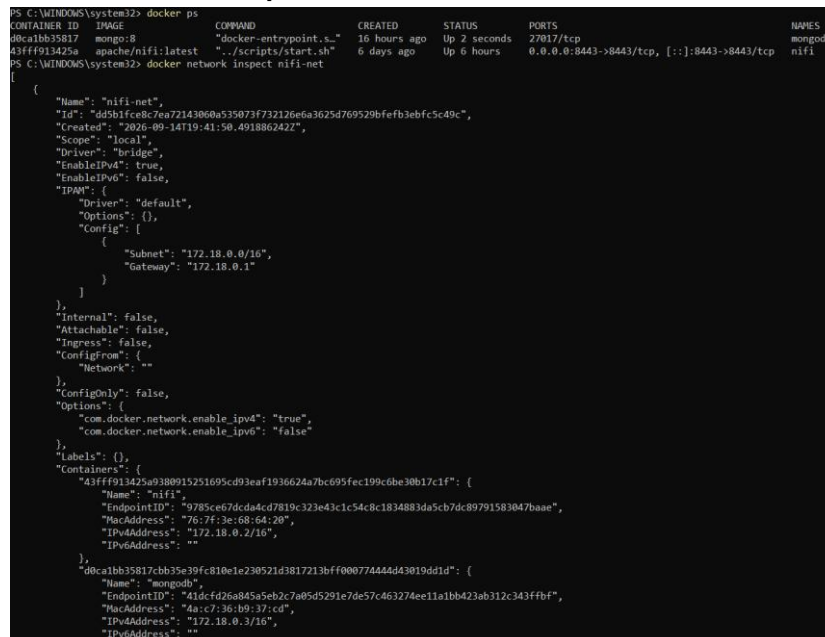
b. Fluxua exekutatu:



5. MongoDB container-a Docker-en:

a. Powershell-en:

- Sortu Docker sare bat:
docker network create nifi-net
- NiFi container-a sare horretara gehitu:
docker network connect nifi-net nifi
- MongoDB container-a sortu:
docker run --name mongodb --network nifi-net -d mongo:8
- Egiaztatu biak martxan daudela:
docker ps
docker network inspect nifi-net



- NiFi-ra itzuli.

6. AttributesToJSON procesadorea:

	AttributesToJSON AttributesToJSON 2.11.0 org.apache.nifi - nifi-standard-nar	
In	16 (908 bytes)	5 min
Read/Write	0 bytes / 208 bytes	5 min
Out	16 (208 bytes)	5 min
Tasks/Time	16 / 00:00:00.013	5 min

Edit Processor | AttributesToJSON 2.11.0

Settings	Scheduling	Properties	Relationships	Comments
Required field		+ Verification ☒		
Property	Value			
Attributes List	 datuak	Click the button above to verify this component.		
Attributes Regular Expression	 No value set			
Destination	 flowfile-content			
Include Core Attributes	 false			
Null Value	 false			
JSON Handling Strategy	 Escaped			
Pretty Print	 true			

7. PutMongo procesadorea:

	PutMongo PutMongo 2.11.0 org.apache.nifi - nifi-mongodb-nar	
In	61 (793 bytes)	5 min
Read/Write	793 bytes / 0 bytes	5 min
Out	0 (0 bytes)	5 min
Tasks/Time	61 / 00:00:00.094	5 min

Processor Details | PutMongo 2.11.0

Settings	Scheduling	Properties	Relationships	Comments
Required field		Verification ☒		
Property	Value			
Client Service	 MongoDBControllerService			
Mongo Database Name	 nifi			
Mongo Collection Name	 datuak			
Mode	 insert			
Character Set	 UTF-8			

Processor Details | PutMongo 2.11.0

Settings	Scheduling	Properties	Relationships	Comments
Required field		Verification ☒		
Property	Value			
Client Service	 MongoDBControllerService 			
Mongo Database Name	 nifi Go to Service			

Controller Services
NIFI Flow > NIFI_Flow_03_kasua_2aldaera

Controller Service Details
MongoDBControllerService 2.11.0

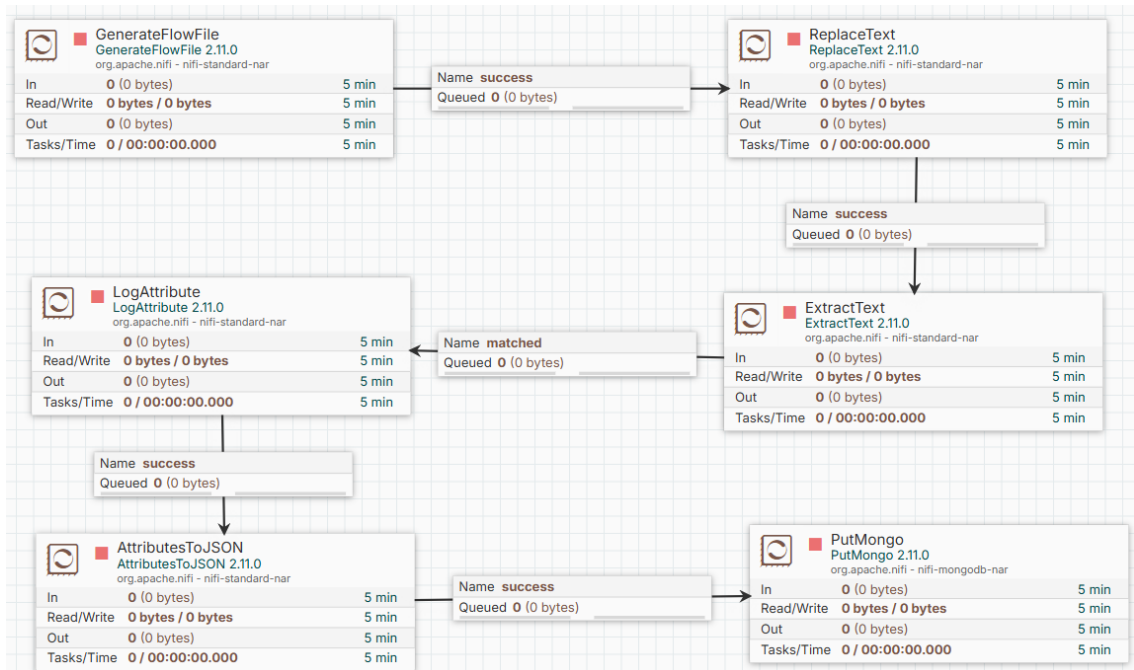
Required field

Property	Value
Mongo URI	 mongodb://mongodb:27017
Database User	 No value set
Password	 No value set
SSL Context Service	 No value set
Write Concern	 ACKNOWLEDGED

Controller Services

NIFI Flow > NIFI_Flow_03_kasua_2aldaera					☐ Show current scope only 
Name ↑	Type	Bundle	State	Scope	
MongoDBControllerService	MongoDBControllerService 2.11.0	org.apache.nifi - nifi-mongodb-services-nar	 Enabled	NIFI_Flow_03_kasua_2aldaera	

8. Fluxu guztia:



9. MongoDB-n egiaztatu:

- a. Powershell-en exekutatu:

docker exec -it mongodb mongosh

```

PS C:\WINDOWS\system32> docker exec -it mongodb mongosh
Current Mongosh Log ID: 6aa8ea341dfb29f93f249a06
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.10.0
Using MongoDB: 8.3.9
Using Mongosh: 2.10.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-09-15T06:35:42.274+00:00: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See
http://dochub.mongodb.org/core/prodnotes-filesystem
2026-09-15T06:35:42.516+00:00: Access control is not enabled for the database. Read and write access to data and conf
figuration is unrestricted
2026-09-15T06:35:42.517+00:00: For customers running the current memory allocator, we suggest changing the contents of
the following sysfsFile
2026-09-15T06:35:42.517+00:00: For customers running the current memory allocator, we suggest changing the contents of
the following sysfsFile
2026-09-15T06:35:42.517+00:00: We suggest setting the contents of sysfsFile to 0.
2026-09-15T06:35:42.517+00:00: vm.max_map_count is too low
2026-09-15T06:35:42.517+00:00: We suggest setting swappiness to 0 or 1, as swapping can cause performance problems.
-----
test>
  
```

- b. Jadanik MongoDB-ren kontsolan gaudela, datu-baseak ikusi:

show dbs

```

test> show dbs
admin 40.00 KiB
config 96.00 KiB
local 72.00 KiB
nifi 88.00 KiB
test>
  
```

- c. Aukeratu nifi datu-basea:

use nifi

```

test> use nifi
switched to db nifi
nifi>
  
```

- d. Ikusi collection-ak:

show collections

```
nifi> show collections
datuak
nifi>
```

e. Ikusi dokumentuak:

db.datuak.find()

```
nifi> db.datuak.find()
[
  {
    _id: ObjectId('6aa8f88ed689e3a10386ea1d'),
    datuak: "a<rbbl(@4#[[ Data: Tue Sep 15 07:49:34 UTC 2026 ]]]'"
  }
]
nifi>
```

Apache NiFi 4. kasua: MongoDB

Helburua: HTTP bidez jasotako testuak prozesatzea, erroreak detektatzea, informazioa metadatu bihurtzea eta MongoDB-n JSON dokumentu gisa gordetzea. Horretarako, NiFi-n datu-fluxu bat sortu behar da. Fluxuak HTTP POST eskaerak jasoko ditu, mezuak ERROR hitza duten edo ez duten arabera sailkatuko ditu, mezuak multzokatu, metadatuak gehitu eta azkenik MongoDB-n gordeko ditu.

Erabiliko ditugun prozesadoreak:

- **ListenHTTP:** HTTP POST mezuak jaso
- **RouteOnContent:** ERROR duten eta gainerako mezuak bereizi
- **MergeContent:** Bide bereko FlowFile-ak elkartu
- **ExtractText + UpdateAttribute:** Edukitik atributuak atera eta metadatuak sortu
- **AttributesToJSON:** Atributuak JSON dokumentu bihurtu
- **PutMongo:** JSON dokumentua MongoDB-n gorde

Kasu honetan, ListenHTTP-k funtziona dezan, 8081 portua ere beharko luke nifi-k. Hasieran sortu genuen container-ean ez genuenez gehitu, arazoak izango ditugu HTTP irakurtzeko. Ariketa hau modu azkar bat, beste container bat sortzea izan daiteke.

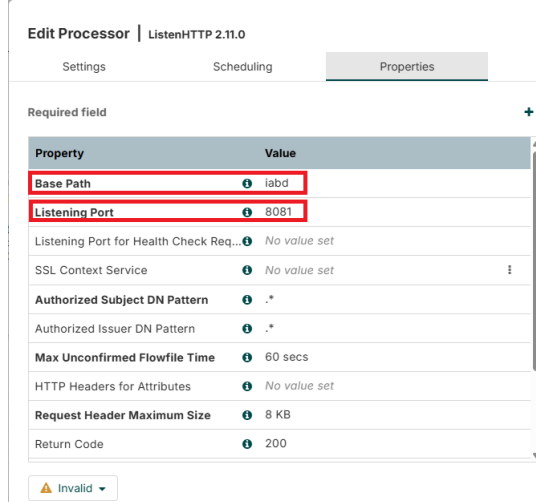
```
docker run --name nifi2 -p 8443:8443 -p 8081:8081 -d -e  
SINGLE_USER_CREDENTIALS_USERNAME=nifi -e  
SINGLE_USER_CREDENTIALS_PASSWORD=nifinifinifi -e NIFI_JVM_HEAP_MAX=2g -v  
"C:\nifi\ariketak:/opt/nifi/ariketak" apache/nifi:latest
```

1. Process Group berria:



2. ListenHTTP prozesadorea:

POST eskaera bakoitzak FF berri bat sortzen du, edukia HTTP body-arekin.



3. RouteOnContent prozesadorea:

Edukiaren arabera bideratzeko. Edukian arakatuko du eta dagokion bideari bidaliko dio (log-iragazketa edo errore-detekziorako oso erabilgarria).

Edit Processor | RouteOnContent 2.11.0

Settings Scheduling **Properties** F

Required field +

Property	Value
Match Requirement	content must contain match
Character Set	UTF-8
Content Buffer Size	1 MB
errorTestua	ERROR

ListenHTTP
ListenHTTP 2.11.0
org.apache.nifi - nifi-standard-nar
In 0 (0 bytes) 5 min
Read/Write 0 bytes / 0 bytes 5 min
Out 0 (0 bytes) 5 min
Tasks/Time 0 / 00:00:00.000 5 min

Name success
Queued 0 (0 bytes)

RouteOnContent
RouteOnContent 2.11.0
org.apache.nifi - nifi-standard-nar
In 0 (0 bytes) 5 min
Read/Write 0 bytes / 0 bytes 5 min
Out 0 (0 bytes) 5 min
Tasks/Time 0 / 00:00:00.000 5 min

4. MergeContent prozesadorea:

FlowFile anitz batuko ditu.

Edit Processor | MergeContent 2.11.0

Settings Scheduling **Properties** F

Required field +

Property	Value
Merge Strategy	Bin-Packing Algorithm
Merge Format	Binary Concatenation
Attribute Strategy	Keep Only Common Attributes
Correlation Attribute Name	RouteOnContent.Route
Minimum Number of Entries	1
Maximum Number of Entries	1000
Minimum Group Size	0 B
Maximum Group Size	No value set
Bin Termination Check	No value set
Max Bin Age	300 s

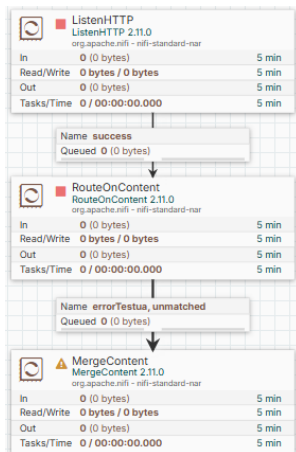
Edit Processor | MergeContent 2.11.0

Settings **Scheduling** Properties

Scheduling Strategy ⓘ*
Timer driven

Concurrent Tasks ⓘ* 1 Run Schedule ⓘ* 30 sec

Execution ⓘ*
All nodes



5. **ExtractText** prozesadorea:

FlowFile-aren edukitik testua atera eta atributu batean gorde.

Edit Processor | ExtractText 2.11.0

Settings Scheduling **Properties** Relationships Comments

Required field + Verification ✓

Property	Value
Enable DOTALL Mode	❗ false
Enable Literal Parsing of the Pattern	❗ false
Enable Multiline Mode	❗ false
Enable Unicode-aware Case Folding	❗ false
Enable Unicode Predefined Chara...	❗ false
Enable Unix Lines Mode	❗ false
Include Capture Group 0	❗ false
Enable Repeating Capture Group	❗ false
Enable Named Groups	❗ false
edukia	❗ (.*)

Click the button above to verify this component.

Invalid Cancel Apply

Edit Processor | MergeContent 2.11.0

Settings Scheduling Properties **Relationships**

Automatically Terminate/Retry Relationships ⓘ

failure

☒ terminate ☐ retry

If the bundle cannot be created, all FlowFiles that would have been used to created the bundle will be transferred to failure

merged

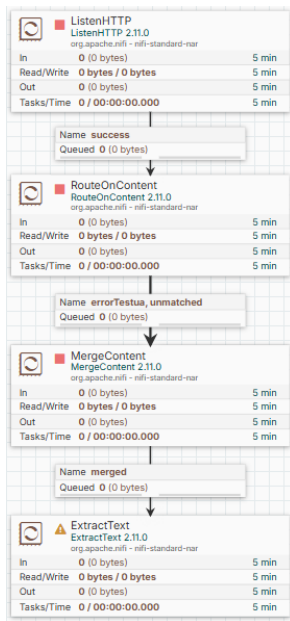
☐ terminate ☐ retry

The FlowFile containing the merged content

original

☒ terminate ☐ retry

The FlowFiles that were used to create the bundle



6. UpdateAttribute prozesadorea: Atributueta metadatuak gehitu.

Edit Processor | UpdateAttribute 2.11.0

Settings Scheduling Properties R

Required field +

Property	Value
Delete Attributes Expression	No value set
Store State	Do not store state
Stateful Variables Initial Value	No value set
Cache Value Lookup Cache Size	100
egoera	\$(RouteOnContent.Route)

Edit Processor | ExtractText 2.11.0

Settings Scheduling Properties Relationships

Automatically Terminate/Retry Relationships

matched

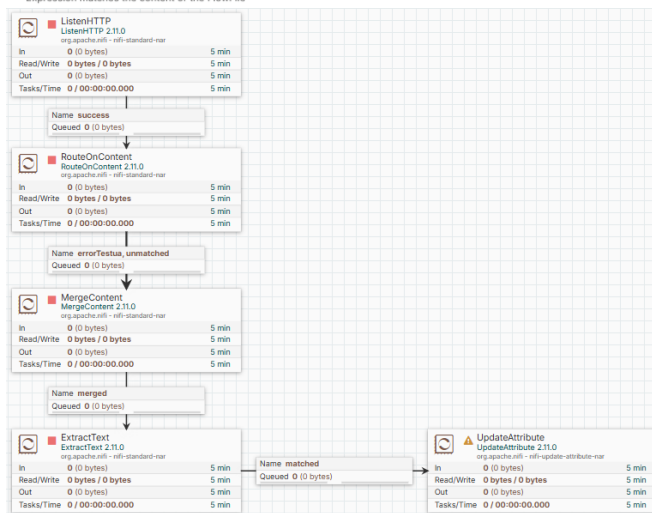
☐ terminate ☐ retry

FlowFiles are routed to this relationship when the Regular Expression is successfully evaluated and the FlowFile is modified as a result

unmatched

☒ terminate ☐ retry

FlowFiles are routed to this relationship when no provided Regular Expression matches the content of the FlowFile



7. **AttributesToJSON** prozesadorea:

PutMongo prozeadoreak FlowFile edukia JSON gisa interpretatzen du.

Edit Processor | AttributesToJSON 2.11.0

Settings Scheduling **Properties** R

Required field +

Property	Value
Attributes List	edukia,egoera
Attributes Regular Expression	No value set
Destination	flowfile-content
Include Core Attributes	true
Null Value	false
JSON Handling Strategy	Escaped
Pretty Print	false

Edit Processor | AttributesToJSON 2.11.0

Settings Scheduling Properties **Relationships**

Automatically Terminate/Retry Relationships

failure

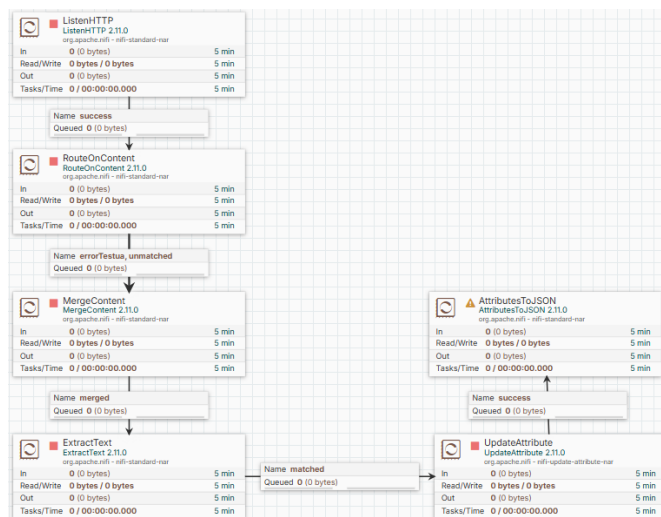
☒ terminate ☐ retry

Failed to convert attributes to JSON

success

☐ terminate ☐ retry

Successfully converted attributes to JSON



8. **PutMongo** prozesadorea:

FlowFile-aren edukia MongoDB-ra bidaltzen du, eta edukia dokumentu gisa gordetzen du konfiguraturako datu-base eta bilduman.

Edit Processor | PutMongo 2.11.0

Settings Scheduling Properties **Relationships** Comments

Required field + Verification

Property	Value
Client Service	MongoDBControllerService
Mongo Database Name	No value set
Mongo Collection Name	No value set

Click the button above to verify this component.

[Go to Service](#)

[Create new service](#)

Edit Controller Service

Property	Value
Mongo URI	mongodb://mongodb:27017
Database User	No value set
Password	No value set
SSL Context Service	No value set
Write Concern	ACKNOWLEDGED

Controller Services

NiFi Flow » NiFi_Flow_04_kasua

☐ Show current scope only

Name ↑	Type	Bundle	State	Scope
MongoDBControllerService	MongoDBControllerSe...	org.apache.nifi - nifi-...	Disabled	NiFi_Flow_04_kasua
				Edit Enable

Edit Processor | PutMongo 2.11.0

Property	Value
Client Service	MongoDBControllerService
Mongo Database Name	iabd
Mongo Collection Name	4kasua
Mode	insert
Character Set	UTF-8

Edit Processor | PutMongo 2.11.0

Settings	Scheduling	Properties	Relationships
----------	------------	------------	---------------

Automatically Terminate/Retry Relationships

failure

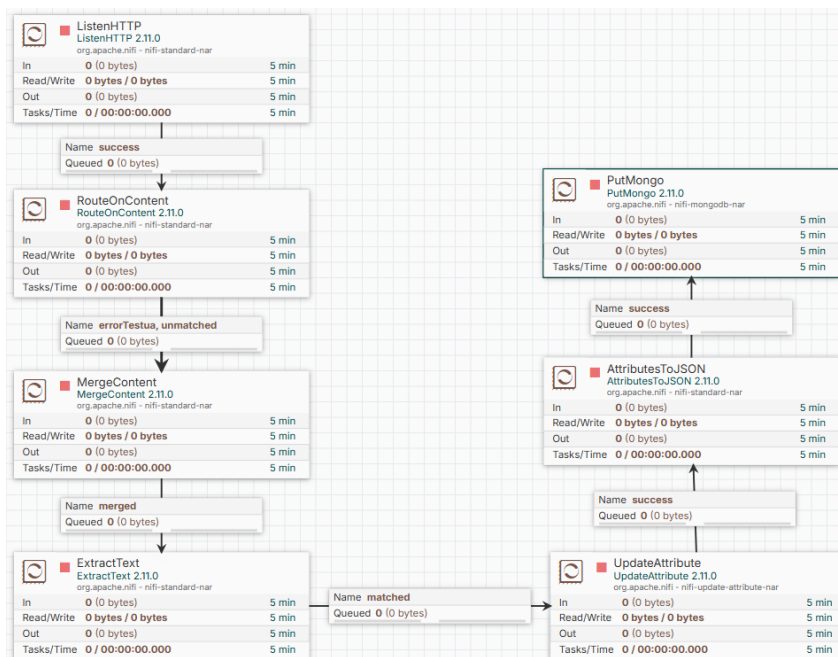
☒ terminate ☐ retry

All FlowFiles that cannot be written to MongoDB are routed to this relationship

success

☐ terminate ☐ retry

All FlowFiles that are written to MongoDB are routed to this relationship



9. Probatzeko:

- a. Probako mezuak bidali. Horretarako, terminalean exekutatu:

```
curl.exe --data "ERROR: zerbitzaria ez dago erabilgarri" http://localhost:8081/iabd
curl.exe --data "Erabiltzaile berria sortu da" http://localhost:8081/iabd
curl.exe --data "ERROR: datu-base konexioak huts egin du" http://localhost:8081/iabd
```

```
PS C:\WINDOWS\System32> curl.exe --data "ERROR: zerbitzaria ez dago erabilgarri" http://localhost:8081/iabd
curl: (52) Empty reply from server
PS C:\WINDOWS\System32> curl.exe --data "ERROR: zerbitzaria ez dago erabilgarri" http://localhost:8081/iabd
PS C:\WINDOWS\System32> curl.exe --data "Erabiltzaile berria sortu da" http://localhost:8081/iabd
PS C:\WINDOWS\System32> curl.exe --data "ERROR: datu-base konexioak huts egin du" http://localhost:8081/iabd
PS C:\WINDOWS\System32> |
```

- b. Egiaztatu MongoDB-n. Horretarako, terminalean:

docker exec -it mongodb mongosh

```
PS C:\WINDOWS\System32> docker exec -it mongodb mongosh
Current Mongosh Log ID: 6aaa3bcbcd47a28954f8fd95
Connecting to:
  mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.10.0
Using MongoDB:
  8.3.9
Using Mongosh:
  2.10.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-09-16T03:55:56.747+00:00: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
2026-09-16T03:55:57.002+00:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2026-09-16T03:55:57.002+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2026-09-16T03:55:57.002+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2026-09-16T03:55:57.002+00:00: We suggest setting the contents of sysfsFile to 0.
2026-09-16T03:55:57.002+00:00: vm.max_map_count is too low
2026-09-16T03:55:57.002+00:00: We suggest setting swappiness to 0 or 1, as swapping can cause performance problems.
-----
test> |
```

Bistaratu datu-baseak:

show dbs

```
test> show dbs
admin    40.00 KiB
config   96.00 KiB
iabd     40.00 KiB
local    80.00 KiB
nifi     68.00 KiB
test> |
```

Aukeratu nifi datu-basea:

use iabd

```
test> use iabd
switched to db iabd
iabd> |
```

Ikusi collection-ak:

show collections

```
iabd> show collections
4kasua
iabd> |
```

Ikusi dokumentuak:

db.getCollection("4kasua").find()

```
iabd> db.getCollection("4kasua").find()
[
  {
    _id: ObjectId('6aaa3ab945105b97fb978890'),
    edukia: 'ERROR: zerbitzaria ez dago erabilgarriERROR: datu-base konexioak huts egin du',
    egoera: 'errorTestua'
  },
  {
    _id: ObjectId('6aaa3ab945105b97fb978891'),
    edukia: 'Erabiltzaile berria sortu da',
    egoera: 'unmatched'
  },
]
```